



Agilent EEsof EDA

Accurate Modeling of Spiral Inductors on Silicon using Planar EM Simulation

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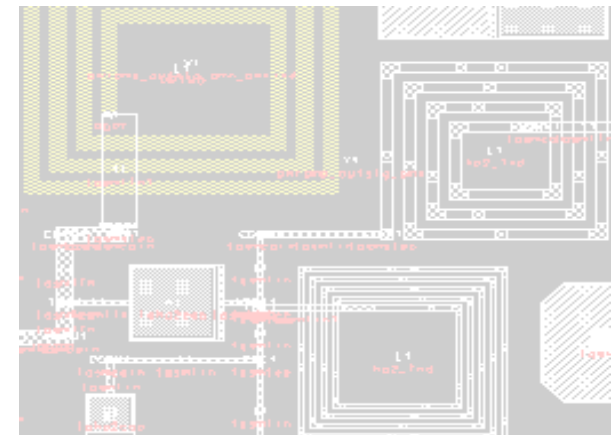
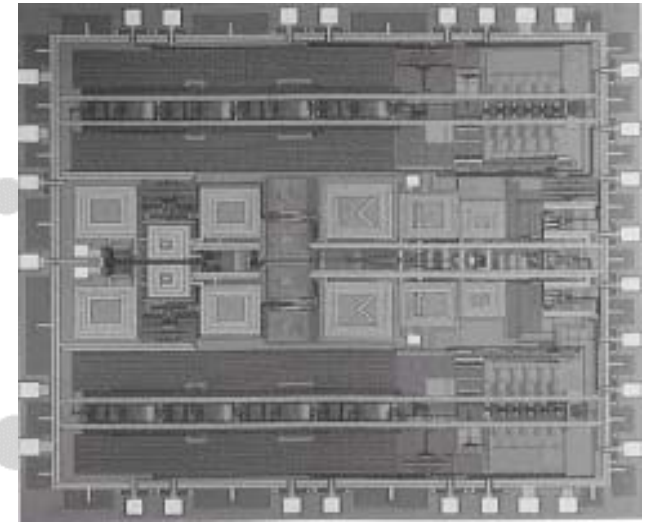
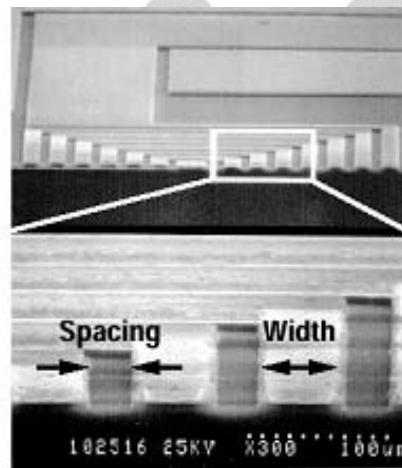
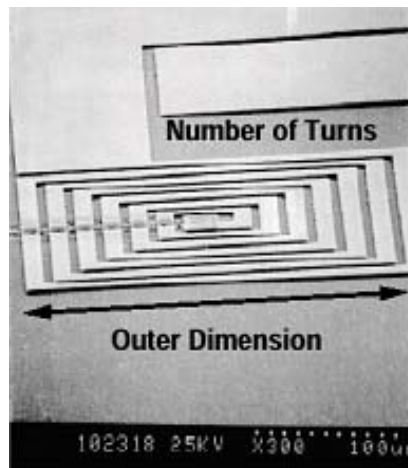
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Accurate Modeling of Spiral Inductors on Silicon From Within Cadence Virtuoso using Planar EM Simulation

Agilent EEsof RFIC Seminar
Spring 2004



Agilent Technologies

Overview

- **Spiral Inductor Models Availability & Limitations**
- **Momentum Technology Overview**
- **Key Physical Effects Considered in Momentum**
- **RFDE Momentum Solution Process**
- **Momentum Application Benchmark**
- **Summary**



Spiral Inductor Models Availability & Limitations

- Inductor models can be found in most RFIC Design Kits
- Spiral inductors are considered critical components in RFIC design
- Foundry Supplied Design Kit inductor models often suffer from:
 - Limited discrete-parameters set
 - Limited frequency coverage
 - Questionable accuracy outside any design/process variations
 - Static model response to surrounding physical environment



Traditional Characterization Method

Measurement-based Modeling Approach

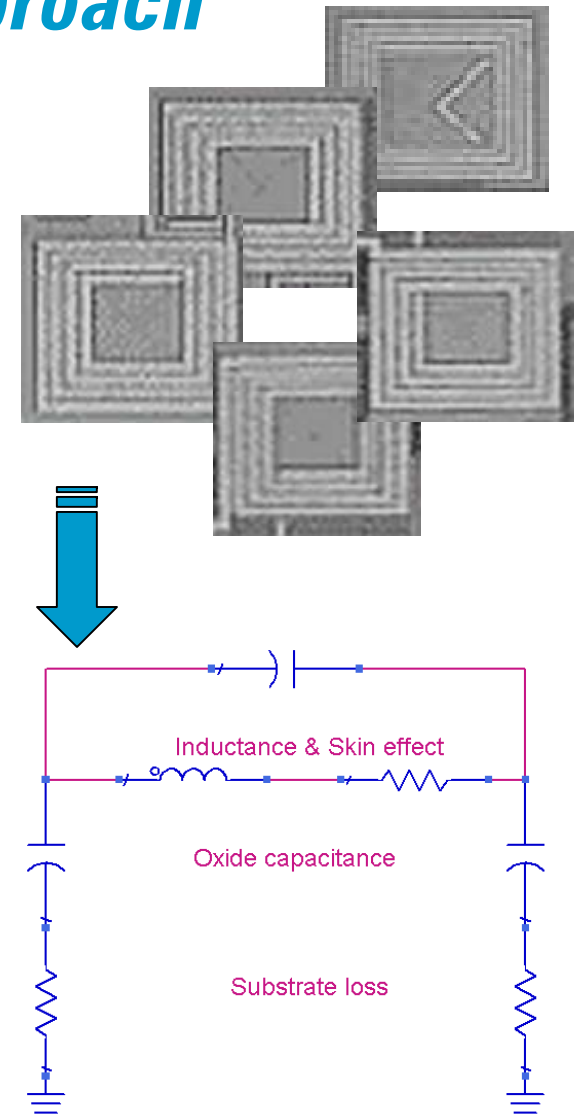
- Fabricate a large number of discretely varying topologies

Positives:

- Good accuracy over predefined frequency range and discrete parameters' samples

Negatives:

- Need to fabricate test wafer(s): Very time consuming and costly
- Limited discrete-parameter set hindering design options
- Static lumped-element models unresponsive to component's physical surroundings
- Need component modeling skills
- Any updates to existing models can be just as costly as starting a new model



Predictive Modeling & Verification

Electromagnetics-based Modeling Approach

- **Perform electromagnetic (EM) simulation**
 - Use Scattering Parameters (S-parameters) data file in frequency-domain circuit simulation
 - Optional: Build lumped element model based on EM data (for time-domain circuit simulation; increased work and possible accuracy impact)

Positives:

- ✓ Minimal EM modeling insight
- ✓ Good accuracy over predefined frequency range and parameter samples
- ✓ Model's frequency range can be extended with few simple menu selections
- ✓ Model's physical parameters can be extended without any need for wafer run
- ✓ Verify model's performance in its intended physical environment

Negative:

- Requires good characterization of process parameters (e.g. substrate)



Overview

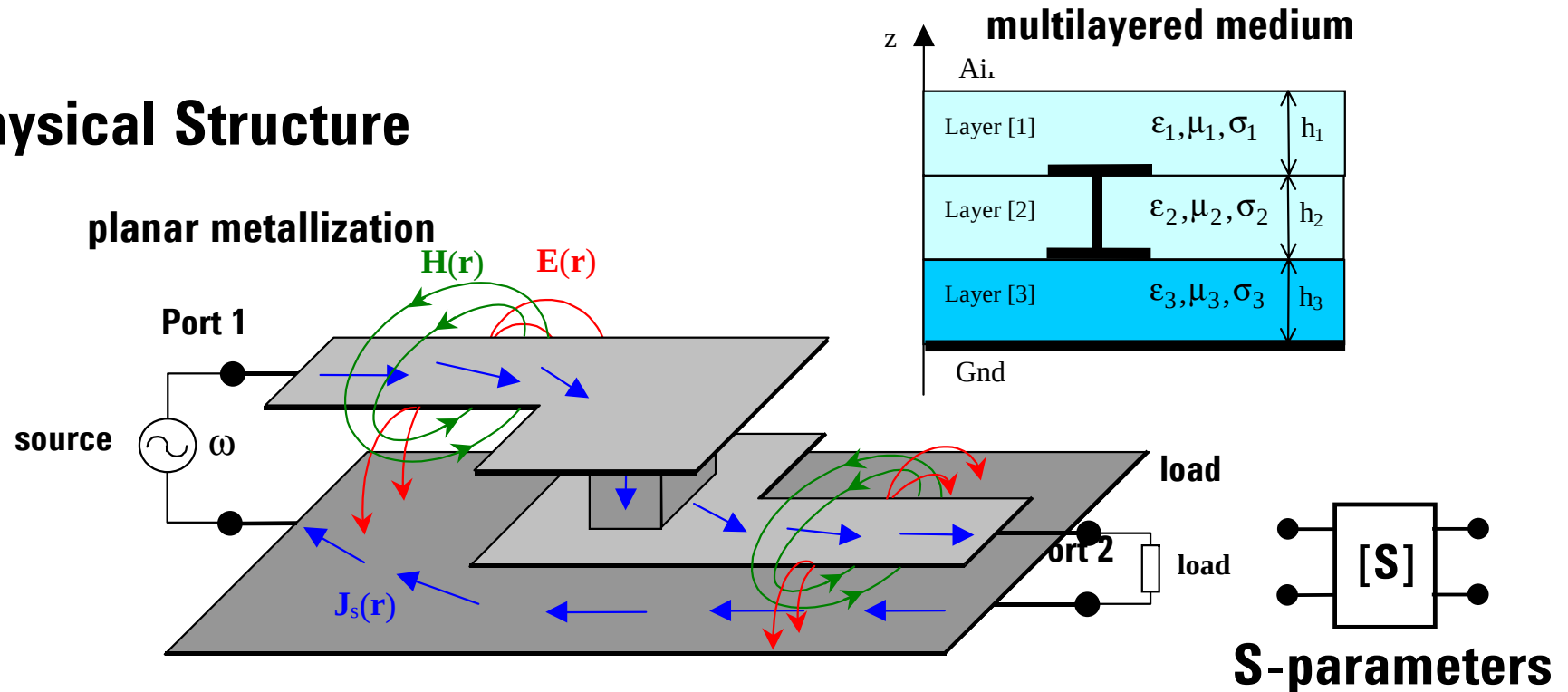
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RFDE Momentum

Electromagnetic Modeling & Verification

Physical Structure

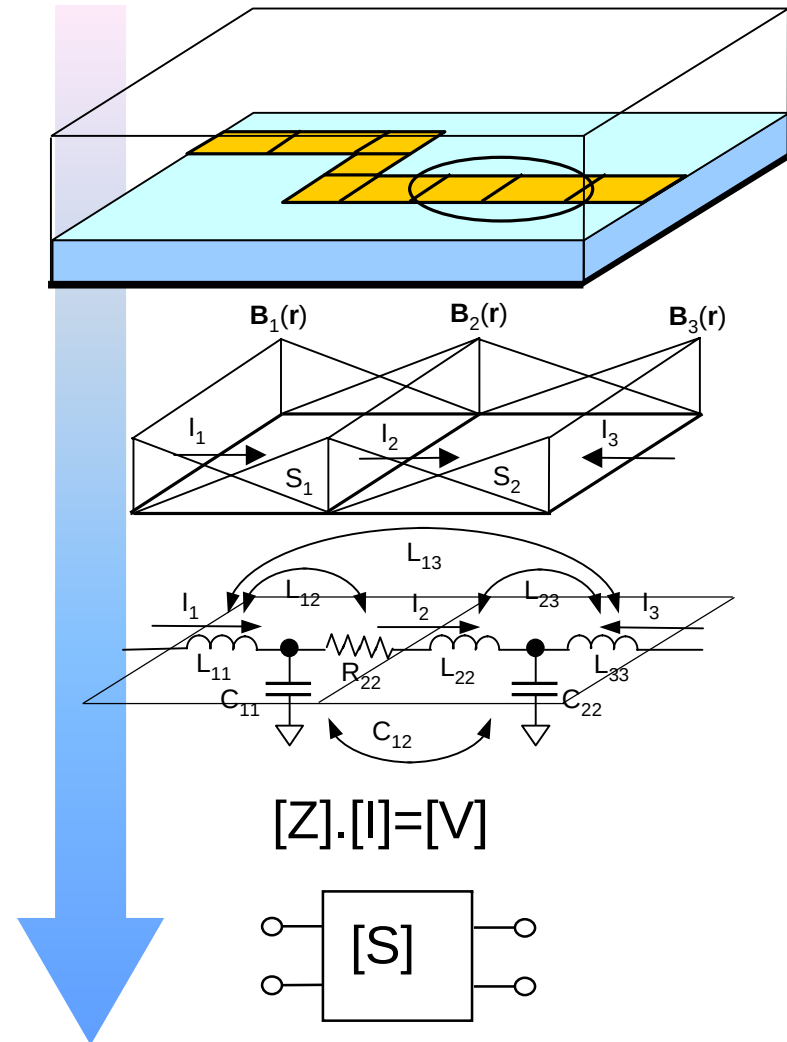


Your "Virtual Network Analyzer"

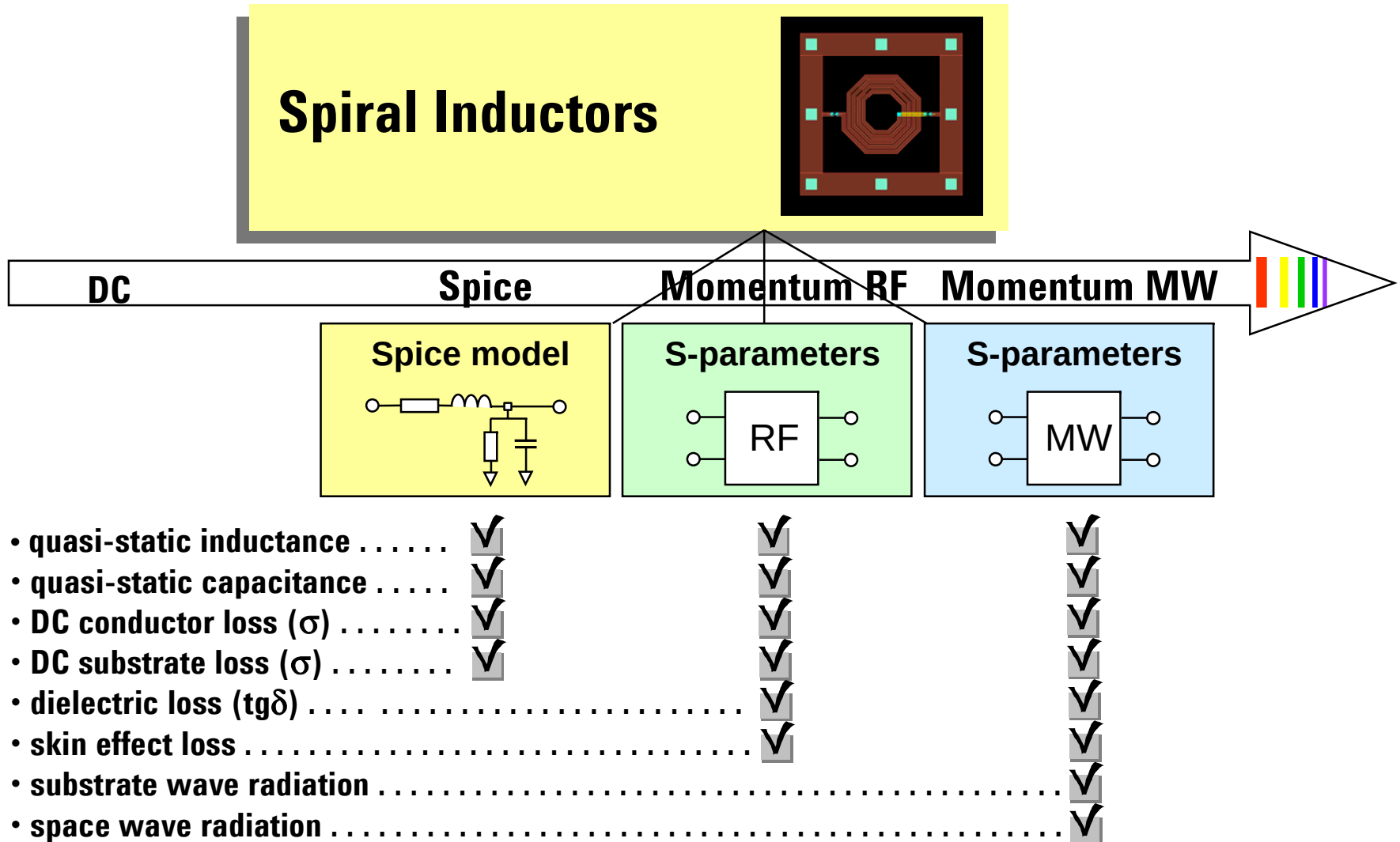
RFDE Momentum

Technology Process

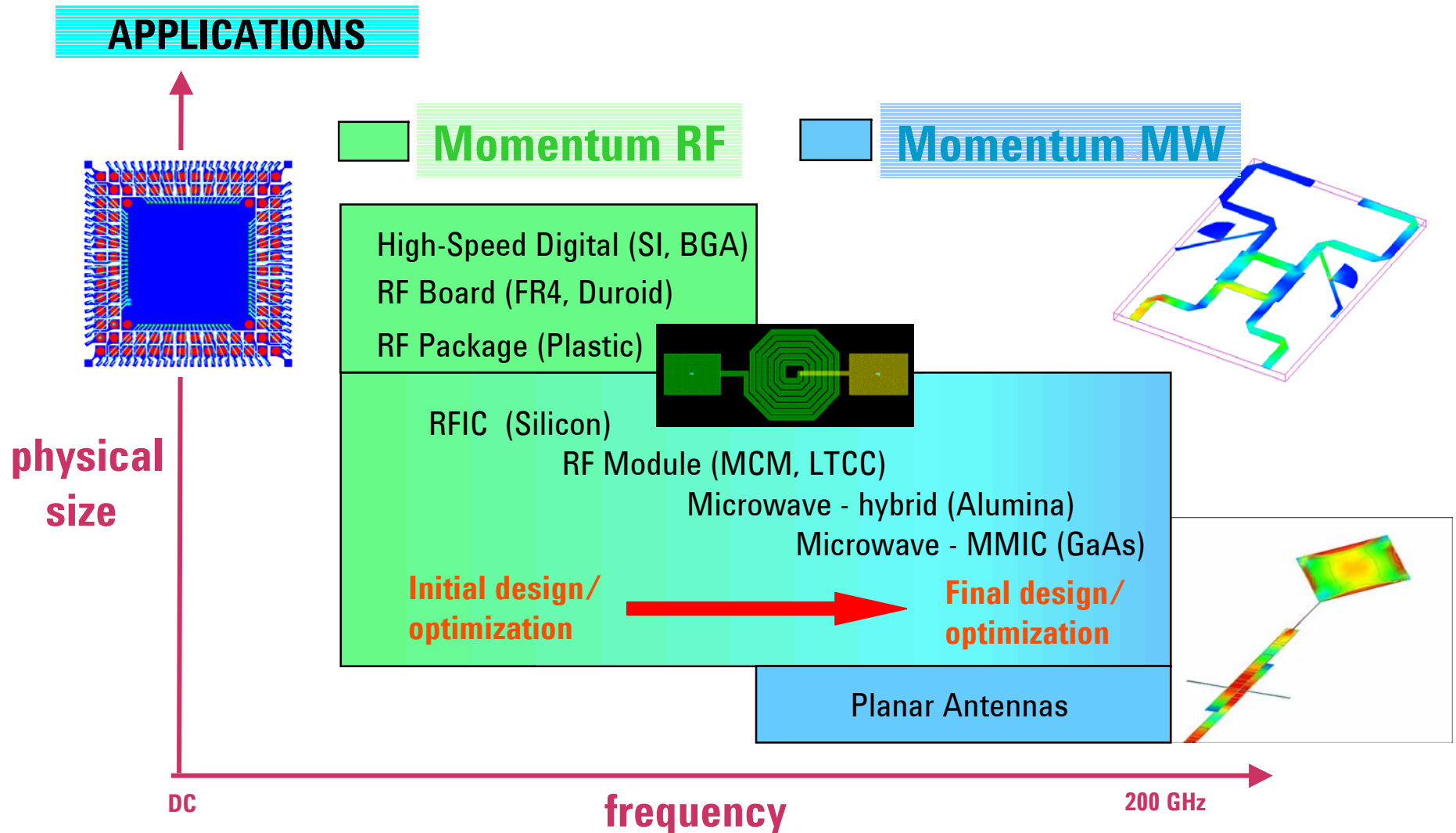
- Calculate Substrate's Green's Function (system's "impulse response")
- Mesh strips, vias and slots with rectangles and triangles (arbitrary surface mesh & mesh refinement)
- Model surface current in each mesh cell (linear distribution)
- Build and solve matrix equation for the unknown current coefficients
- Calculate S-parameters



Comparison of Models

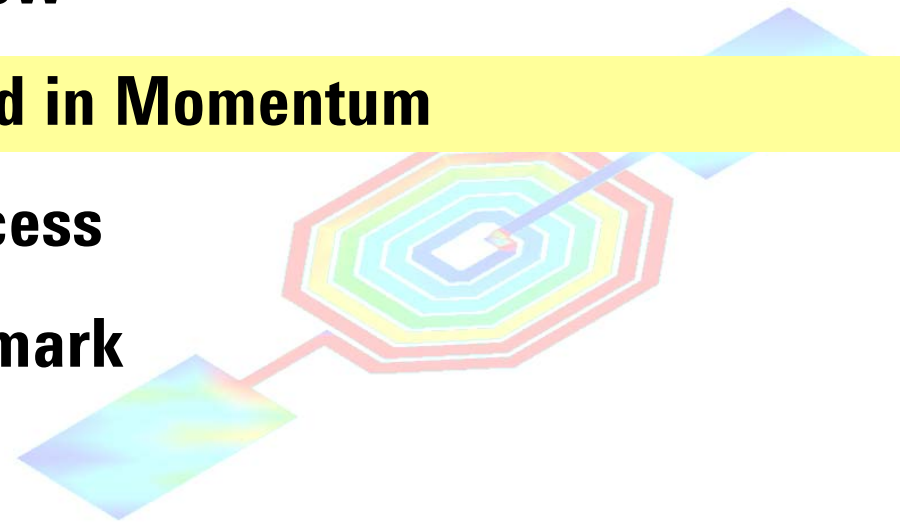


Momentum Microwave & RF – Applications



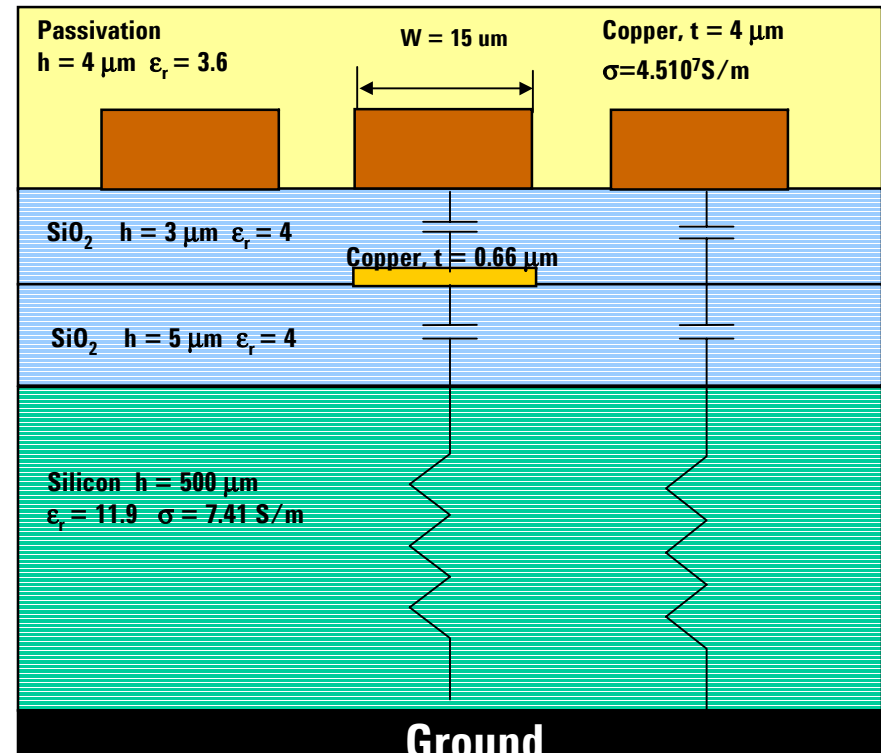
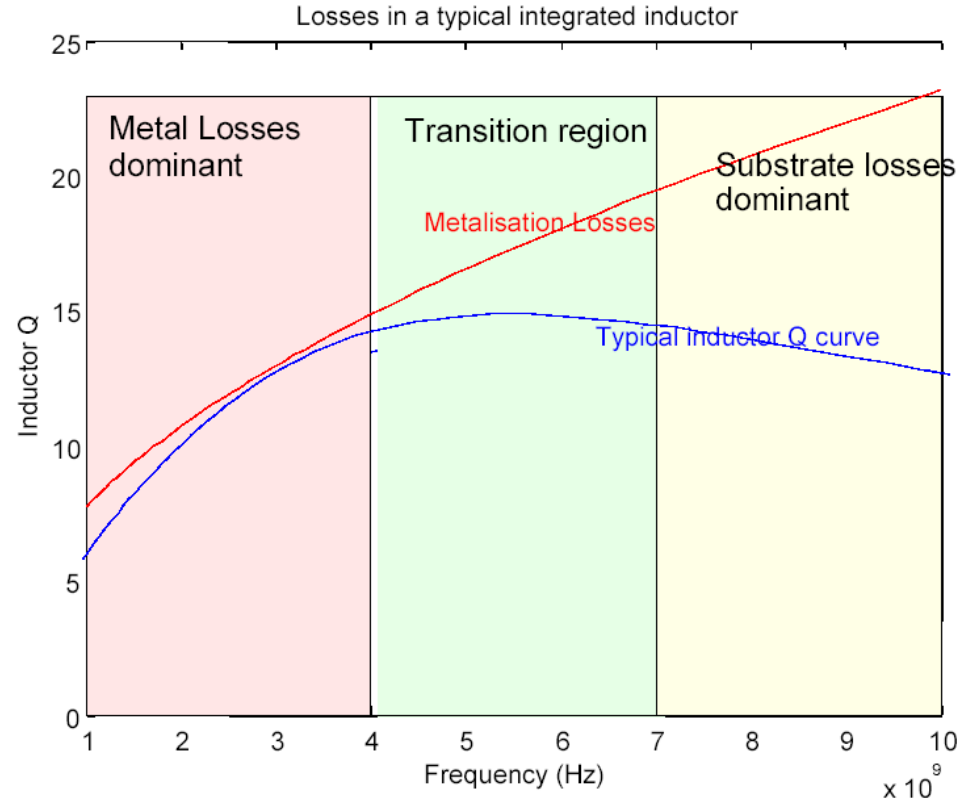
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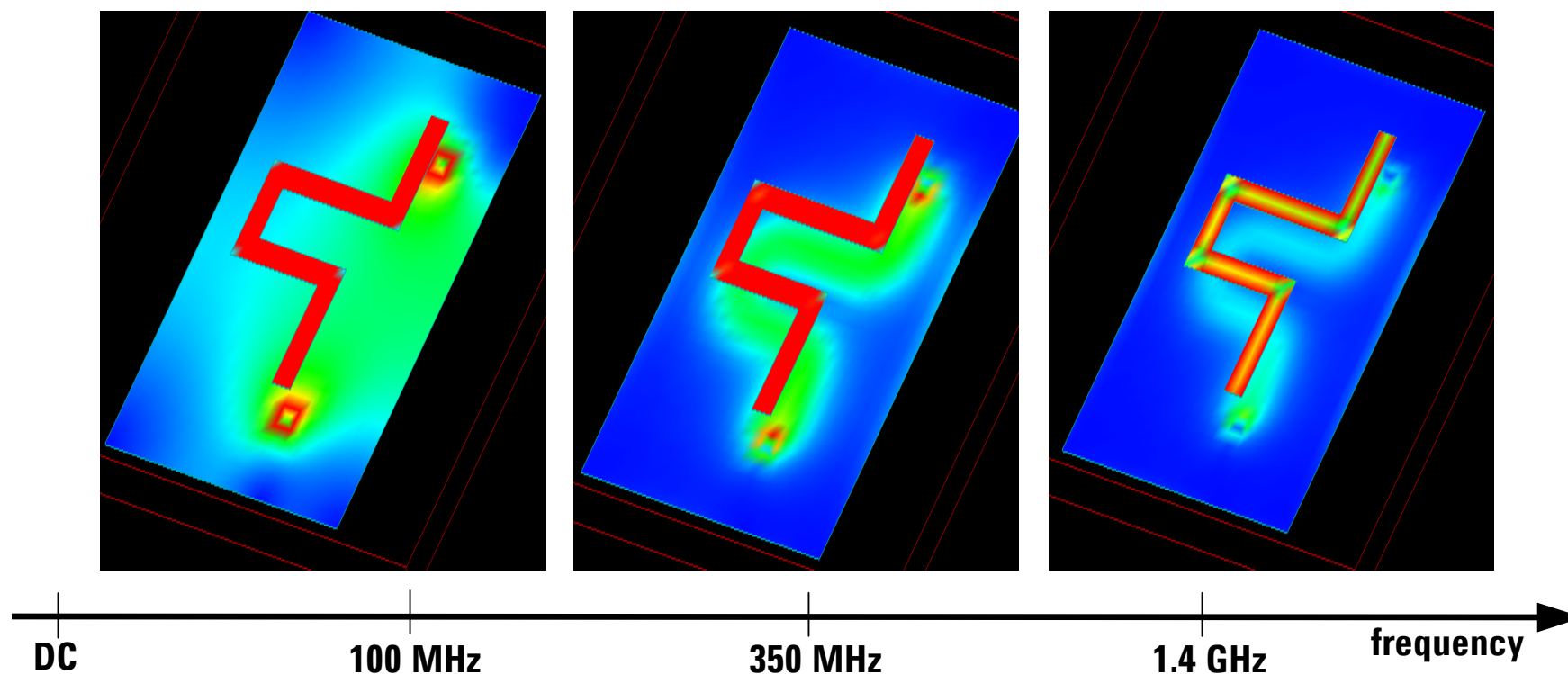


Substrate Coupling & Radiation

- ✓ Resistive loss of metallization
- ✓ Resistivity of Si substrate
- ✓ Capacitive coupling effects to Si substrate



Current Return Path & Frequency Dependency



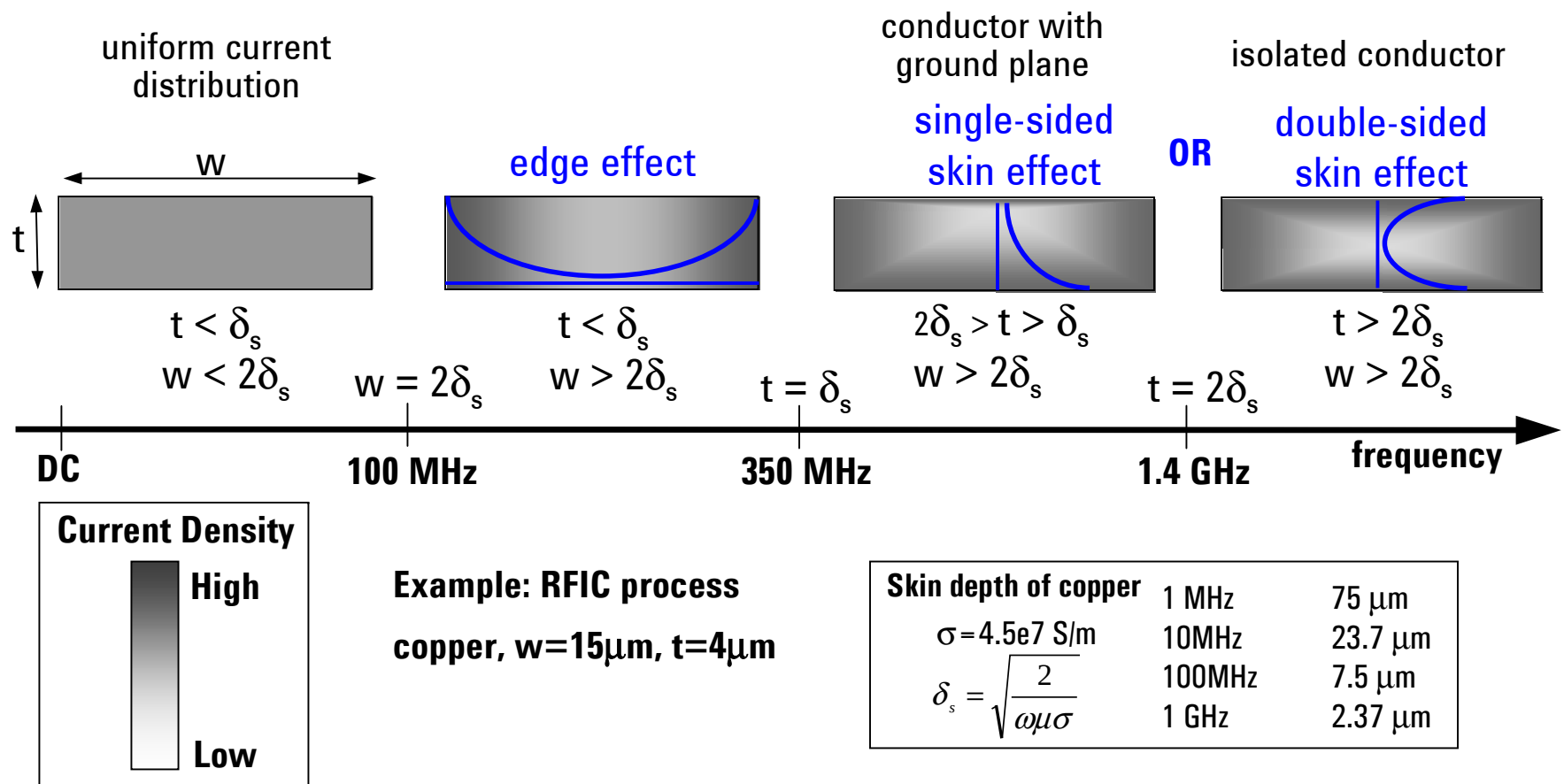
impedance
Current follows the path of “~~least resistance~~”

$$\text{Impedance} = R + j\omega L$$



Current Distribution & Skin Effects

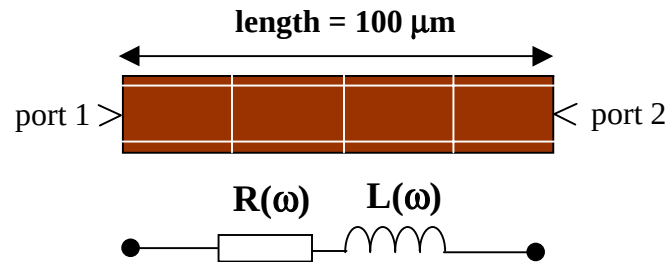
Current distribution in cross section of thick conductor as function of frequency



Thick Conductors Effects

Influence of the **thickness** on the external inductance of the trace

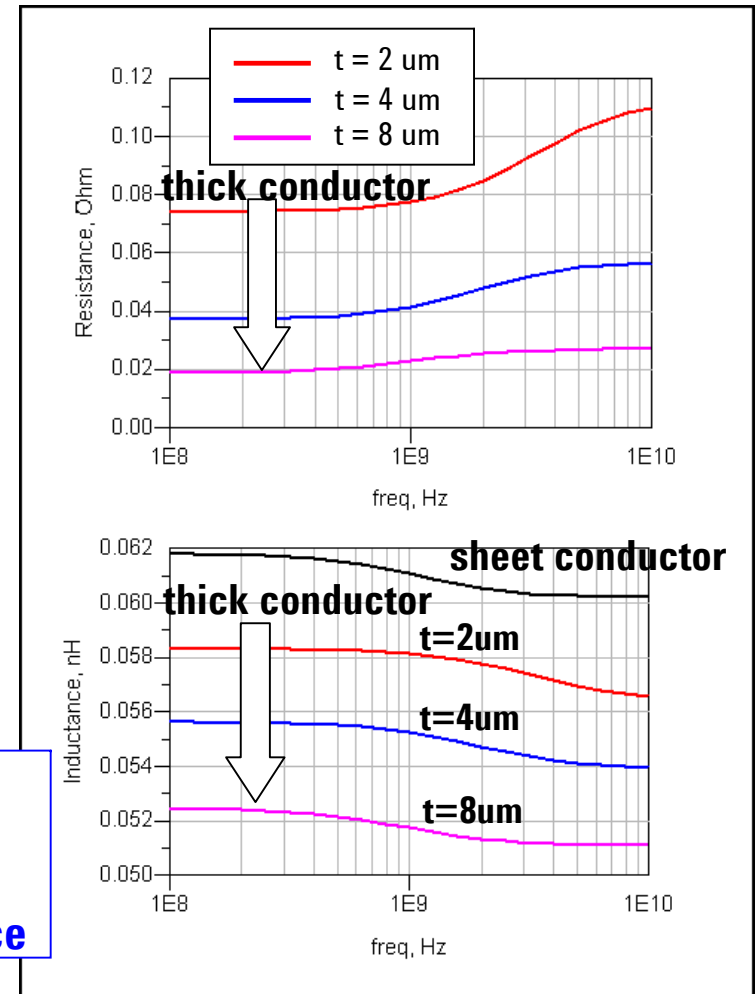
- Using edge mesh to include edge effect
- Using Ohms/square resistance (no skin effect)
- Using thick conductor model



Passivation $h = 4\ \mu\text{m}$ $\epsilon_r = 3.6$	$W = 15\ \mu\text{m}$	Copper, $t = 2, 4, 8\ \mu\text{m}$ $\sigma = 4.510^7\text{S/m}$
SiO_2 $h = 3\ \mu\text{m}$ $\epsilon_r = 4$		Copper, $t = 0.66\ \mu\text{m}$
SiO_2 $h = 5\ \mu\text{m}$ $\epsilon_r = 4$		
Silicon $h = 500\ \mu\text{m}$ $\epsilon_r = 11.9$ $\sigma = 6.67\text{S/m}$		

Increased thickness:

- lower resistance
- lower external inductance



Thick Conductor Modeling Issues

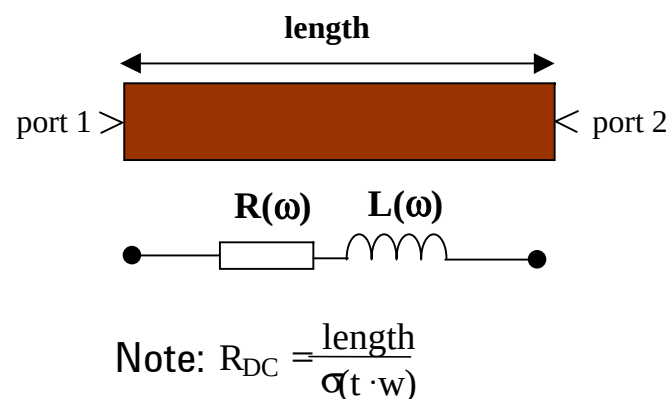
Edge Effect & Skin Effect

How does Momentum address Edge Effect and Skin Effect?

- The **edge mesh** concept takes the **edge effect** into account
- The **surface impedance** concept takes the **skin effect** into account

How does the inclusion of the edge effect and the skin effect influence the resistance and inductance of a trace?

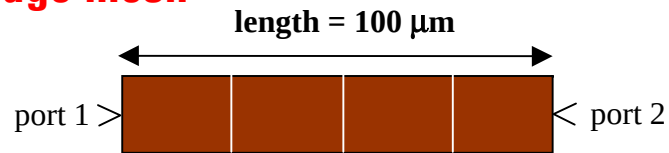
Passivation $h = 4 \mu\text{m}$ $\epsilon_r = 3.6$	$W = 15 \mu\text{m}$	Copper, $t = 4 \mu\text{m}$ $\sigma = 4.510^7 \text{ S/m}$
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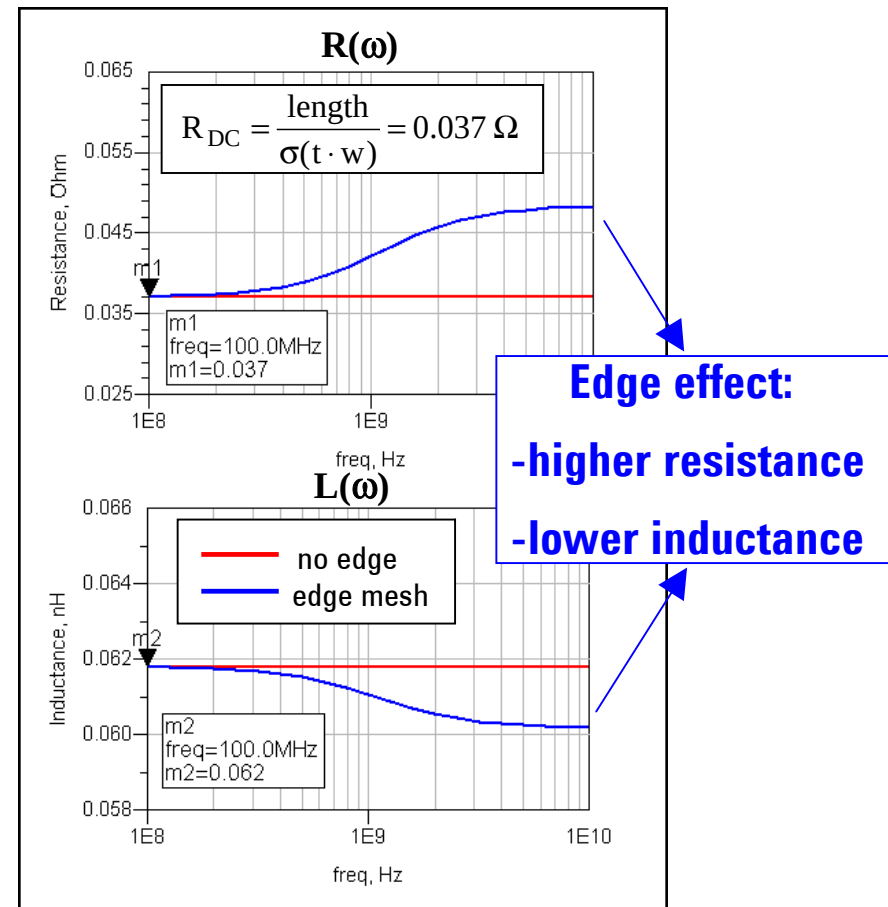
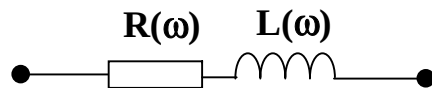
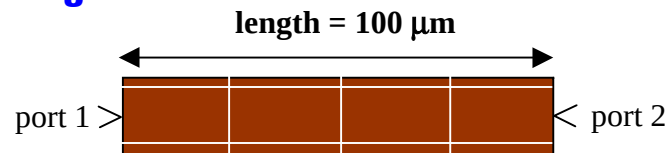
Edge effect

Influence of the **edge effect** on the resistance and inductance of the trace using Ohms/square resistance (no skin effect)

no edge mesh



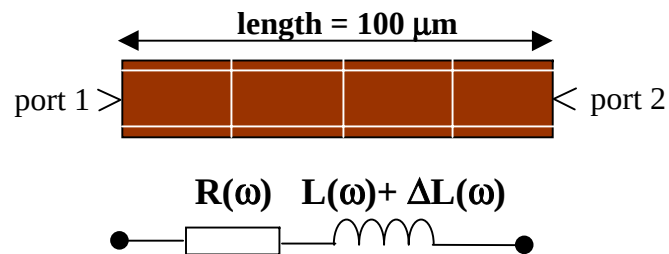
with edge mesh



Skin Effects

Influence of the **skin effect** on the resistance and inductance of a trace

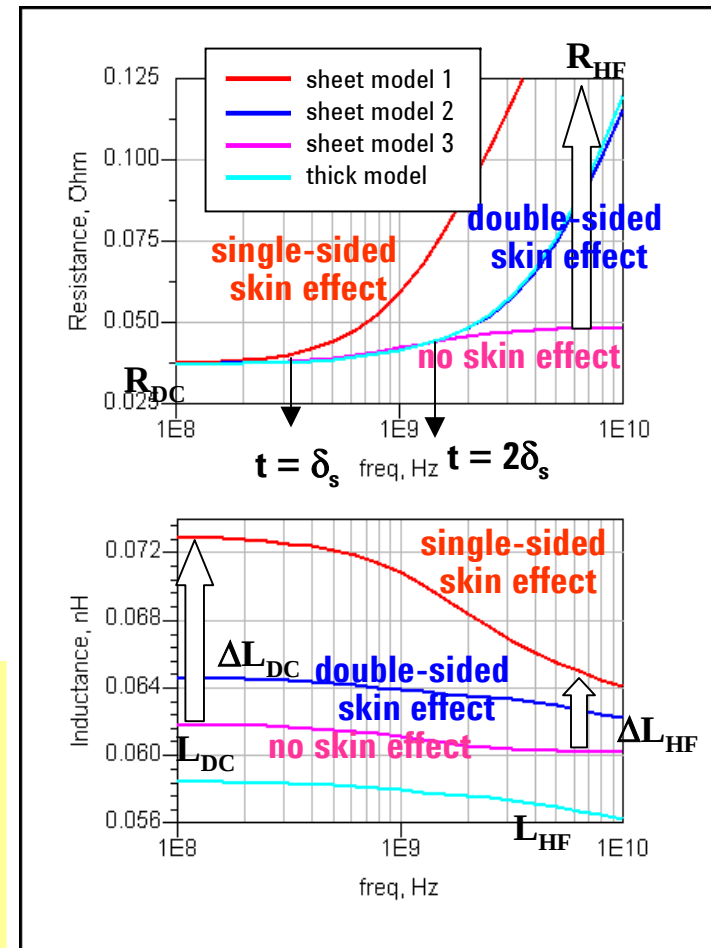
- using edge mesh to include edge effect
- using surface current (sheet) model
- using thick conductor expansion



Passivation $h = 4 \mu\text{m}$ $\epsilon_r = 3.6$	$W = 15 \mu\text{m}$	Copper, $t = 4 \mu\text{m}$
SiO_2 $h = 3 \mu\text{m}$ $\epsilon_r = 4$		Copper, $t = 0.66 \mu\text{m}$
SiO_2 $h = 5 \mu\text{m}$ $\epsilon_r = 4$		
Silicon $h = 500 \mu\text{m}$ $\epsilon_r = 11.9$ $\sigma = 7.41 \text{ S/m}$		

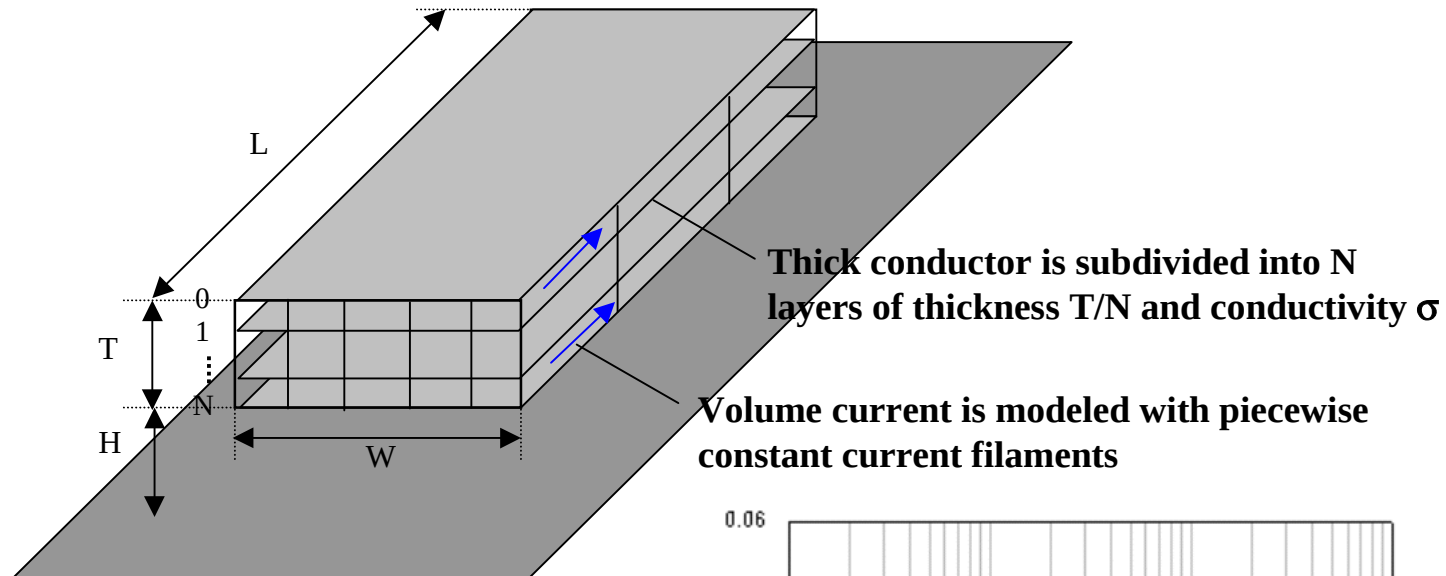
Skin effect:

- higher HF resistance
- higher internal inductance
- lower external inductance



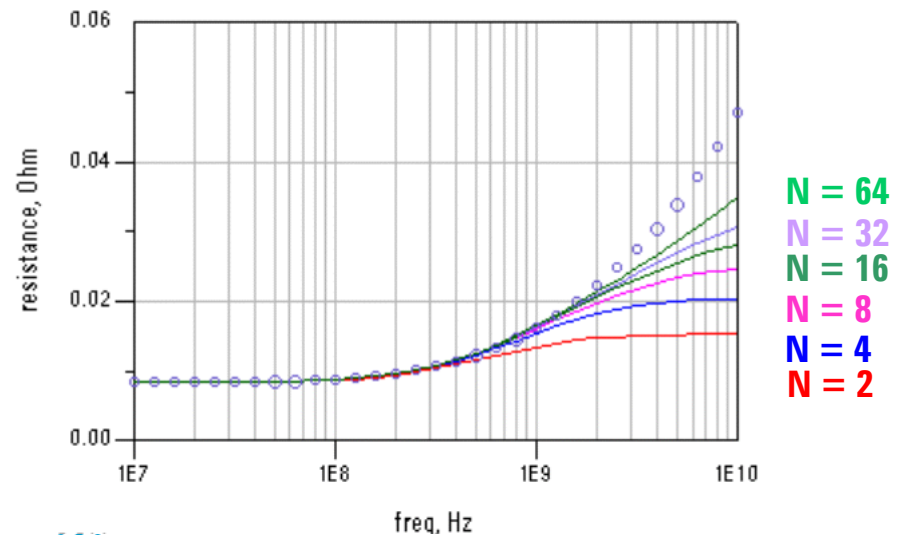
Skin Effect – Discretization Problems

How does a 3D volume meshing technology handles thick conductors?



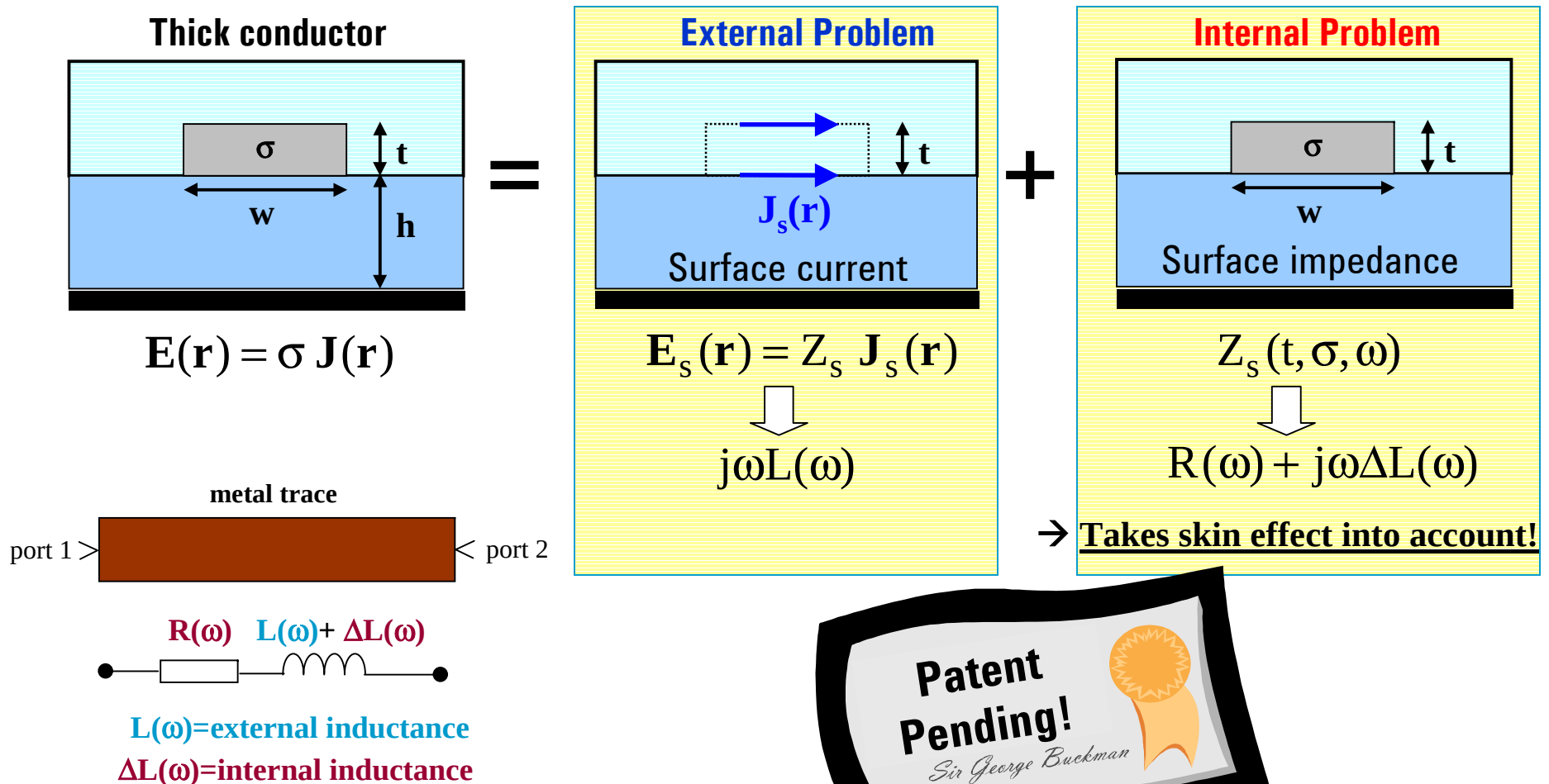
Need at least 3 subdivision per skin depth for convergence !!!

Skin depth of copper		
	1 MHz	75 μm
$\sigma = 4.5\text{e}7 \text{ S/m}$	10MHz	23.7 μm
$\delta_s = \sqrt{\frac{2}{\omega\mu\sigma}}$	100MHz	7.5 μm
	1 GHz	2.37 μm

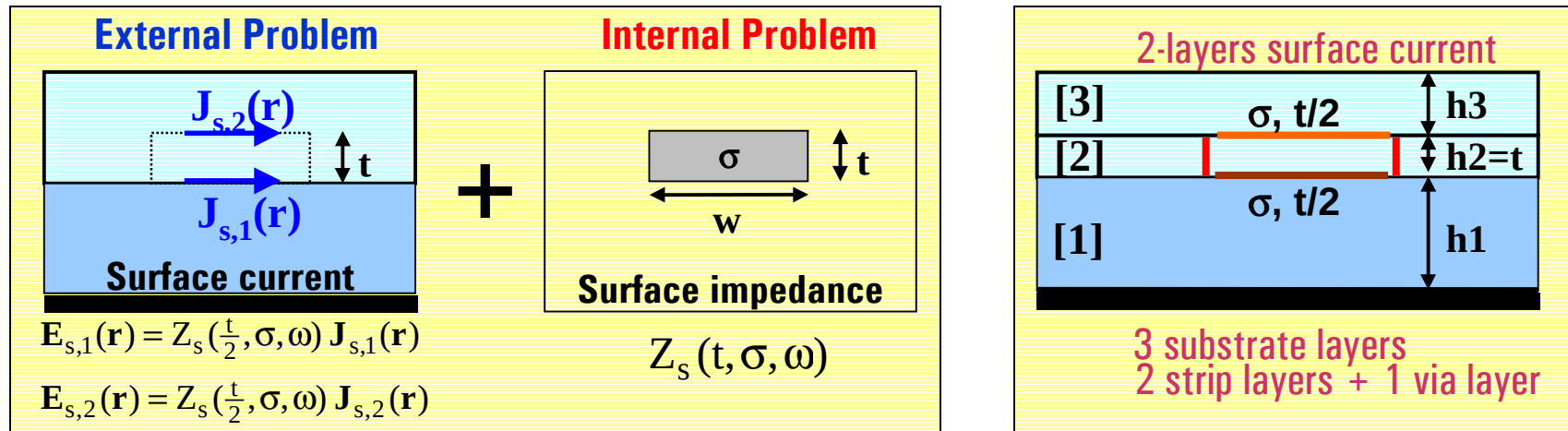


Momentum Modeling of Thick Metal

Field Equivalence Theorem: decomposition into external and internal problem



Thick Conductor Modeling



- Volume of **thick conductor** is divided into two equal **sheet conductor layers**, one at the top surface and one at the bottom surface, with same conductivity and half of the thickness
- Additional **via layer** (perfect conducting) to **short out differential mode**
- Actual distribution of top and bottom layer currents not enforced, but follows from solution of EM equations, yielding **improved model for resistance and inductance**
- **Accuracy decreasing** (horizontal currents on via layers missing) **when w/t decreases** ($w/t < 2$) (this will be addressed in a future release)

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RFDE Momentum

Solution Process Steps

- **Define Substrate stack & layer mapping**
- **Make Momentum cell from design**
- **Map Momentum ports to Cadence pins**
- **Define simulation options: frequency plan, mesh settings...**
- **Perform simulation and review results**
- **Transpose new model to schematic for more accurate circuit simulation**

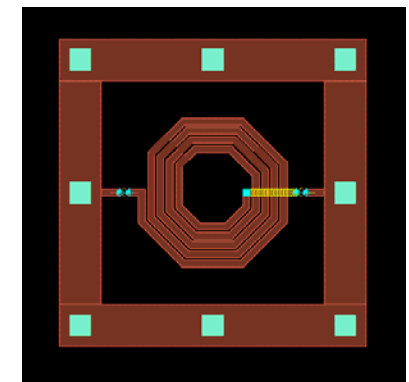
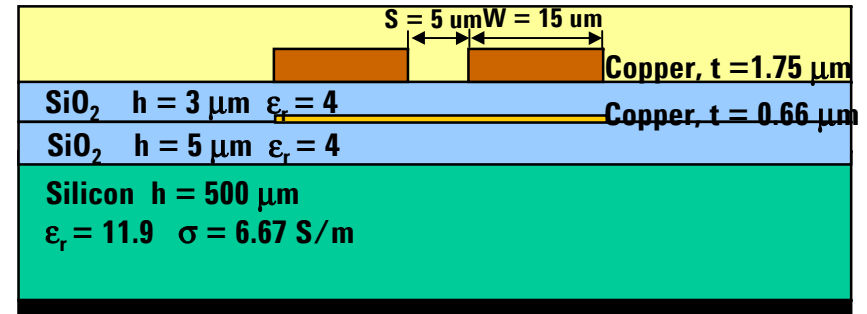


RFDE Momentum Solution Process

Step 1 – Define Substrate stack & layer mapping

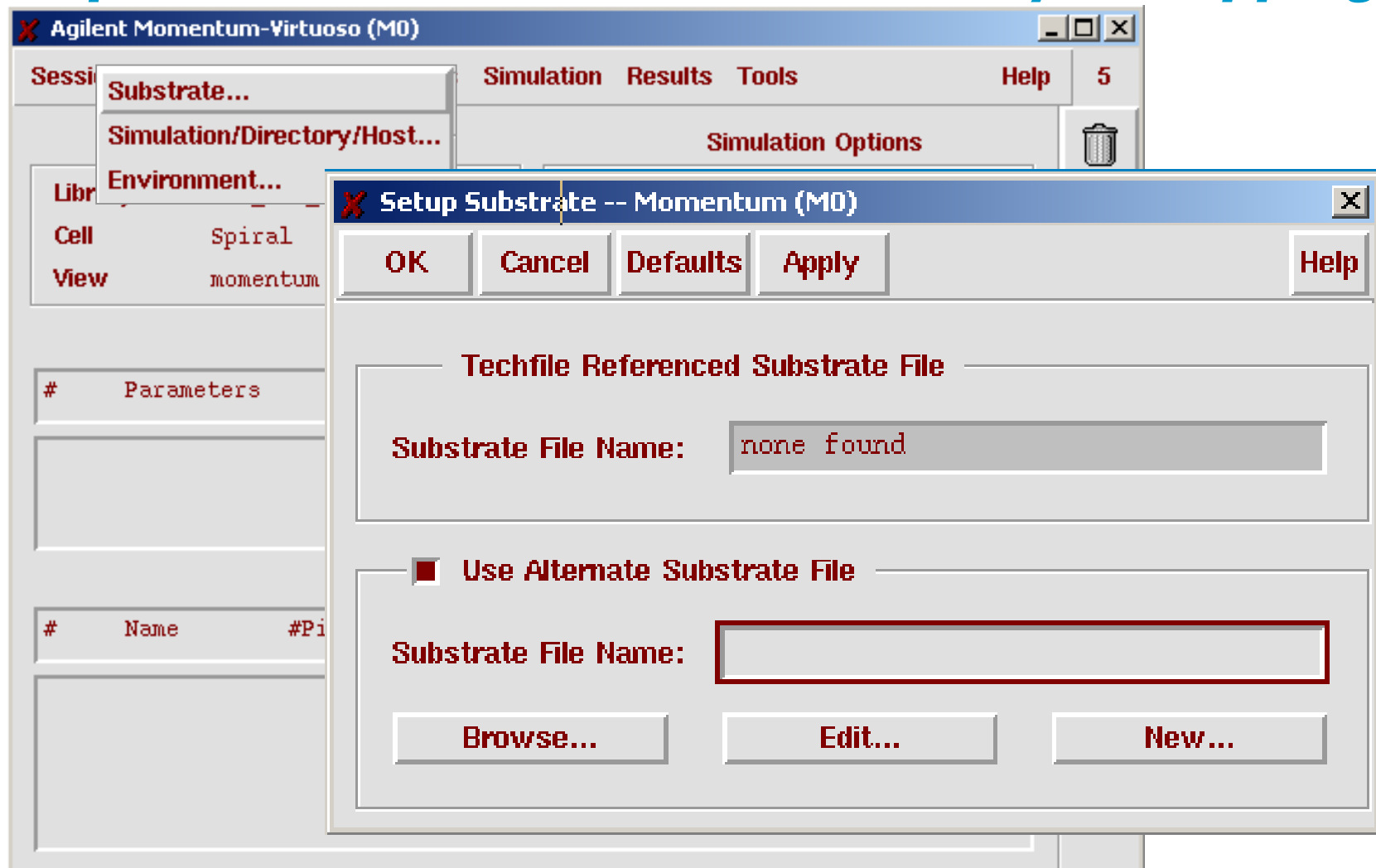
```

UNITS
DISTANCE = M
RESISTANCE = OHM/SQ
CONDUCTIVITY = SIEMENS/M
RESISTIVITY = OHM.CM
PERMEABILITY = RELATIVETO VACUUM
PERMITTIVITY = RELATIVETO VACUUM
END_UNITS
BEGIN_STACK
TOP OPEN PERMITTIVITY = 1 PERMEABILITY = 1
METAL STRIP_MASK = {6}
DIELECTRIC VIA_MASK = {2} HEIGHT = 3e-006 PERMITTIVITY = 4 LOSSTANGENT = 0.001 PERMEABILITY = 1
METAL STRIP_MASK = {1}
DIELECTRIC VIA_MASK = {3} HEIGHT = 5e-006 PERMITTIVITY = 4 LOSSTANGENT = 0.001 PERMEABILITY = 1
METAL ABSENT
DIELECTRIC HEIGHT = 0.0005 PERMITTIVITY = 11.9 RESISTIVITY = 14.9925 PERMEABILITY = 1
BOTTOM COVERED RESISTANCE = 0
END_STACK
BEGIN_MASK
MASK 6 PRECEDENCE = 1 CONDUCTIVITY = 3.77e+007 THICKNESS = 1.75e-006
MASK 1 PRECEDENCE = 0 CONDUCTIVITY = 3.77e+007 THICKNESS = 6.6e-007
MASK 3 PRECEDENCE = 0 CONDUCTIVITY = 3.33e+007
MASK 2 PRECEDENCE = 0 RESISTANCE = 0
END_MASK
    
```



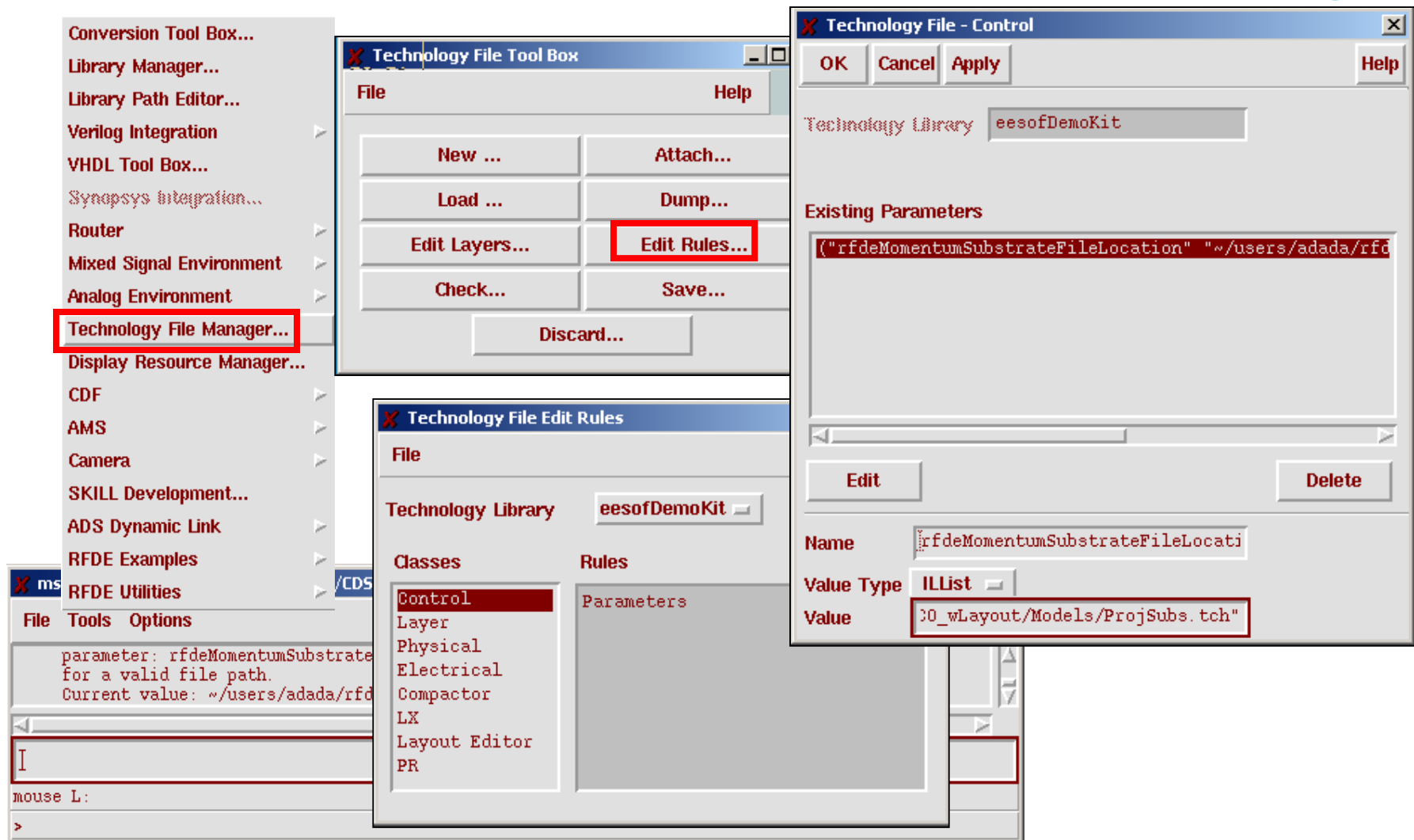
RFDE Momentum Solution Process

Step 1 – Define Substrate stack & layer mapping



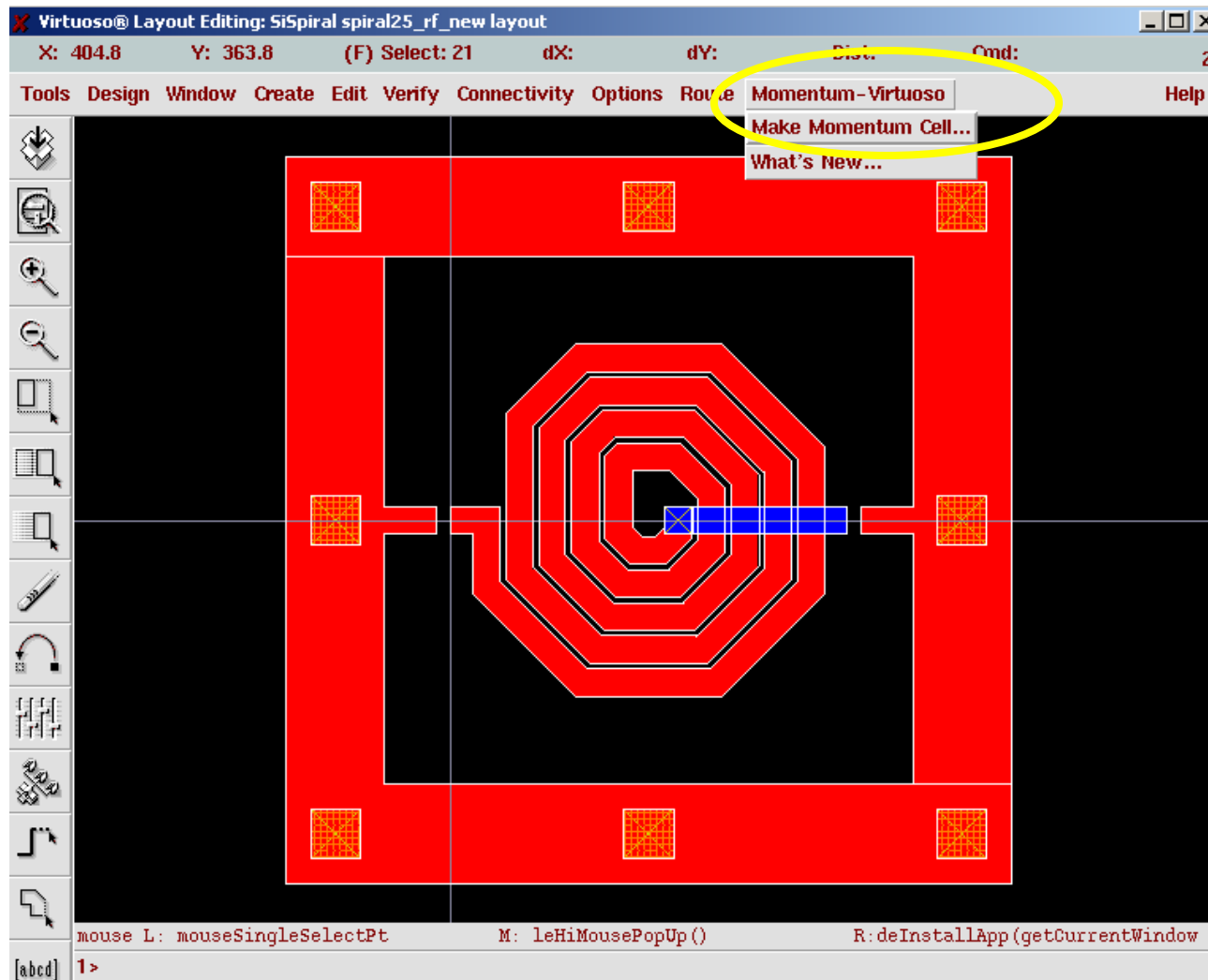
RFDE Momentum Solution Process

Step 1 – Define Substrate stack & layer mapping



RFDE Momentum Solution Process

Step 2 – Make Momentum Cell



RFDE Momentum Solution Process

Step 3 – Assign Ports to Cadence Pins

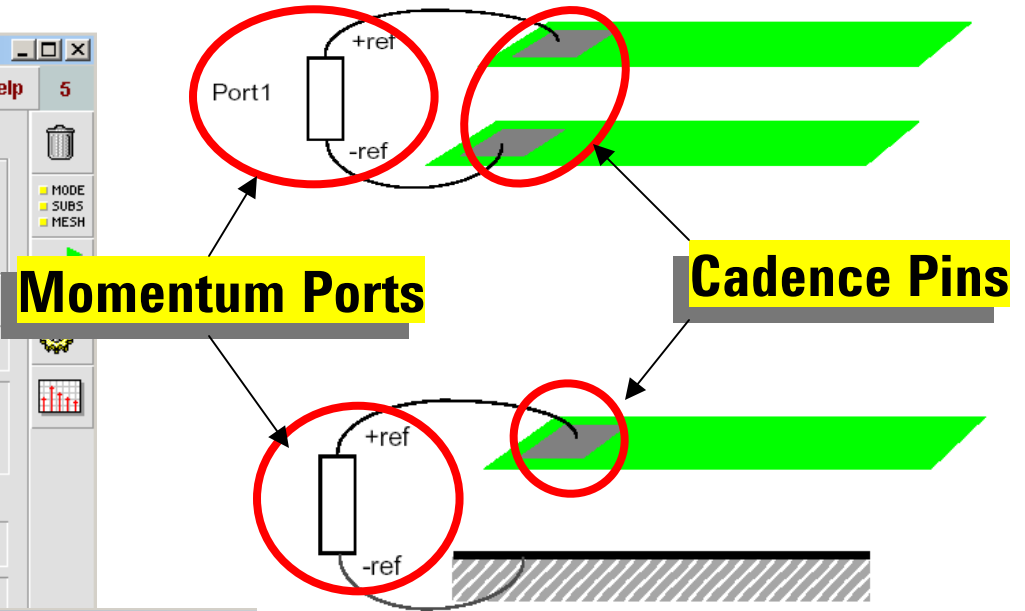
The screenshot shows the Agilent Momentum-Virtuoso (M0) software interface. The 'Ports' tab is selected, displaying a table of existing ports and a dialog box for adding a new port.

Ports							
#	Name	#Pins	Real	Imag	Type	Group	Offset
1	Pin2	1	50.0	0.0			
2	Pin3						
3	Pin4						
4	Pin1						

The 'Add Port -- Momentum (M0)' dialog box is open, showing the following fields:

- Name: Port1
- Impedance: Real (Ohm) 50.0, Imaginary (Ohm) 0.0
- Reference Pins:

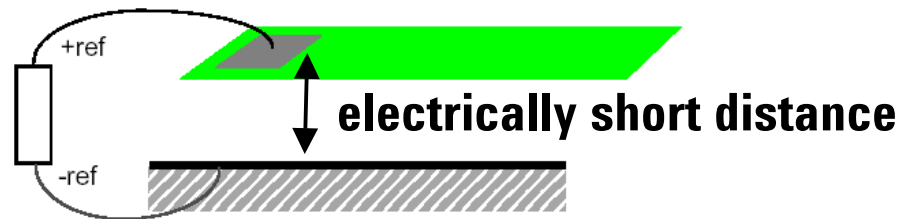
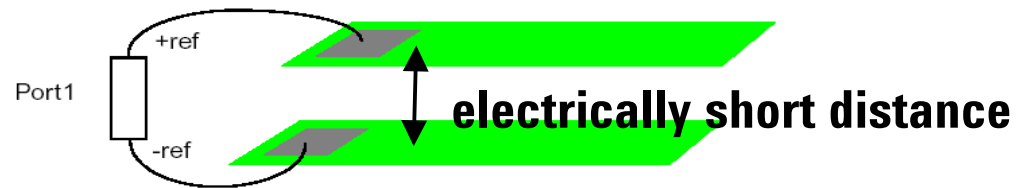
#	Name	Type	Group	Status
1	Pin2			USED
2	Pin3			USED
3	Pin4			USED
4	Pin1			USED



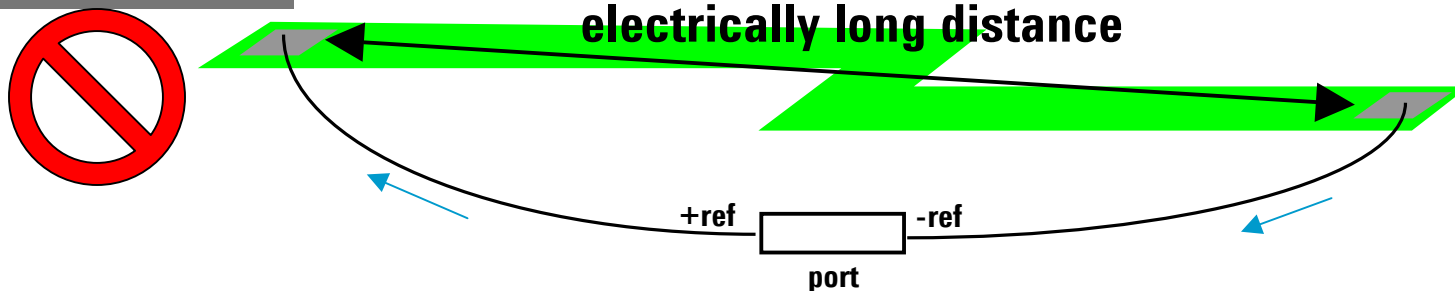
RFDE Momentum Design Flow

Step 3 – Assign Ports to Cadence Pins

Physical ports



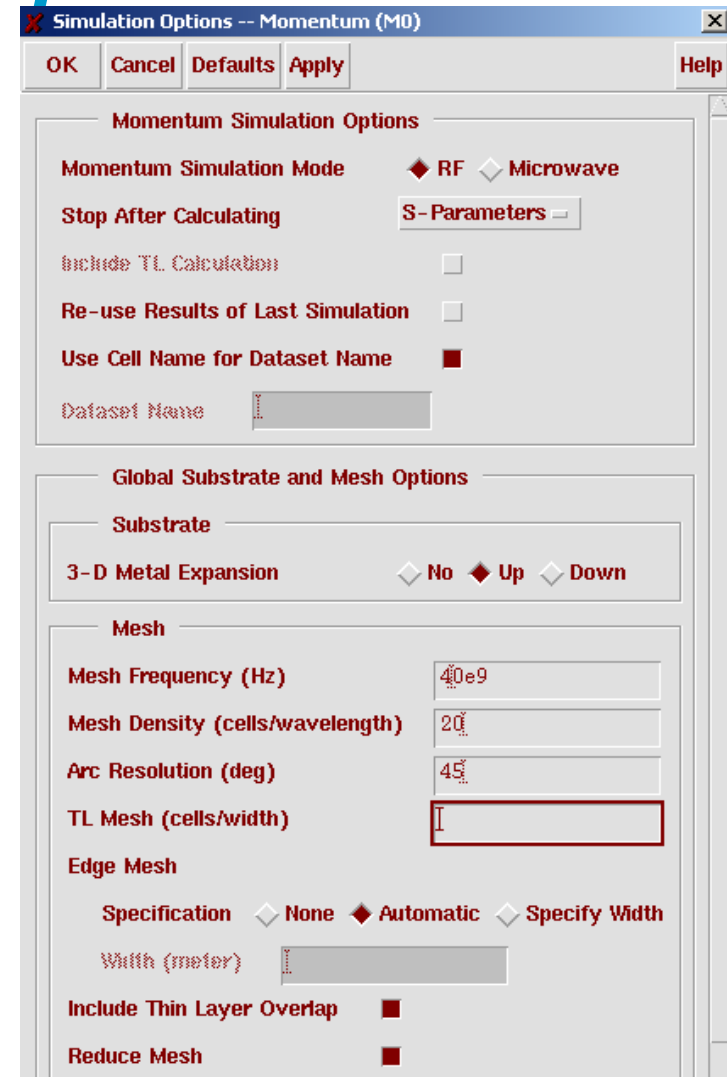
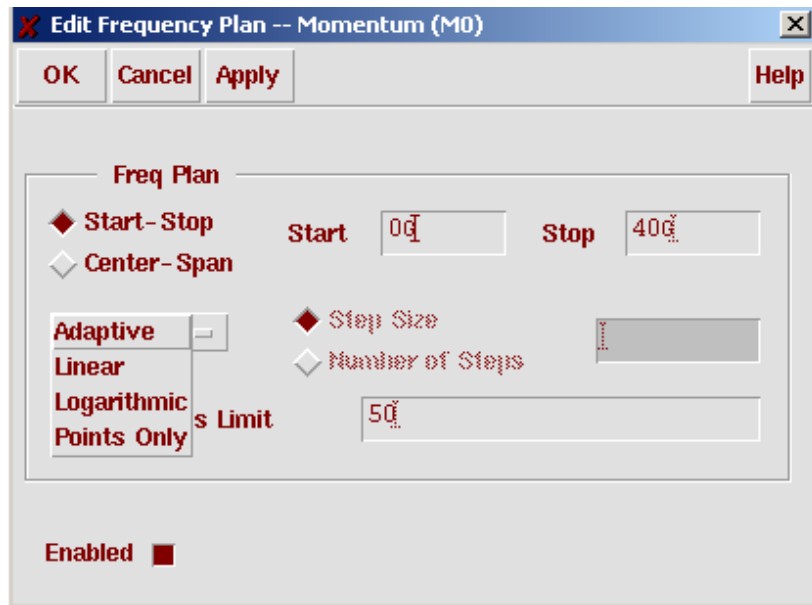
Unphysical ports



RFDE Momentum Solution Process

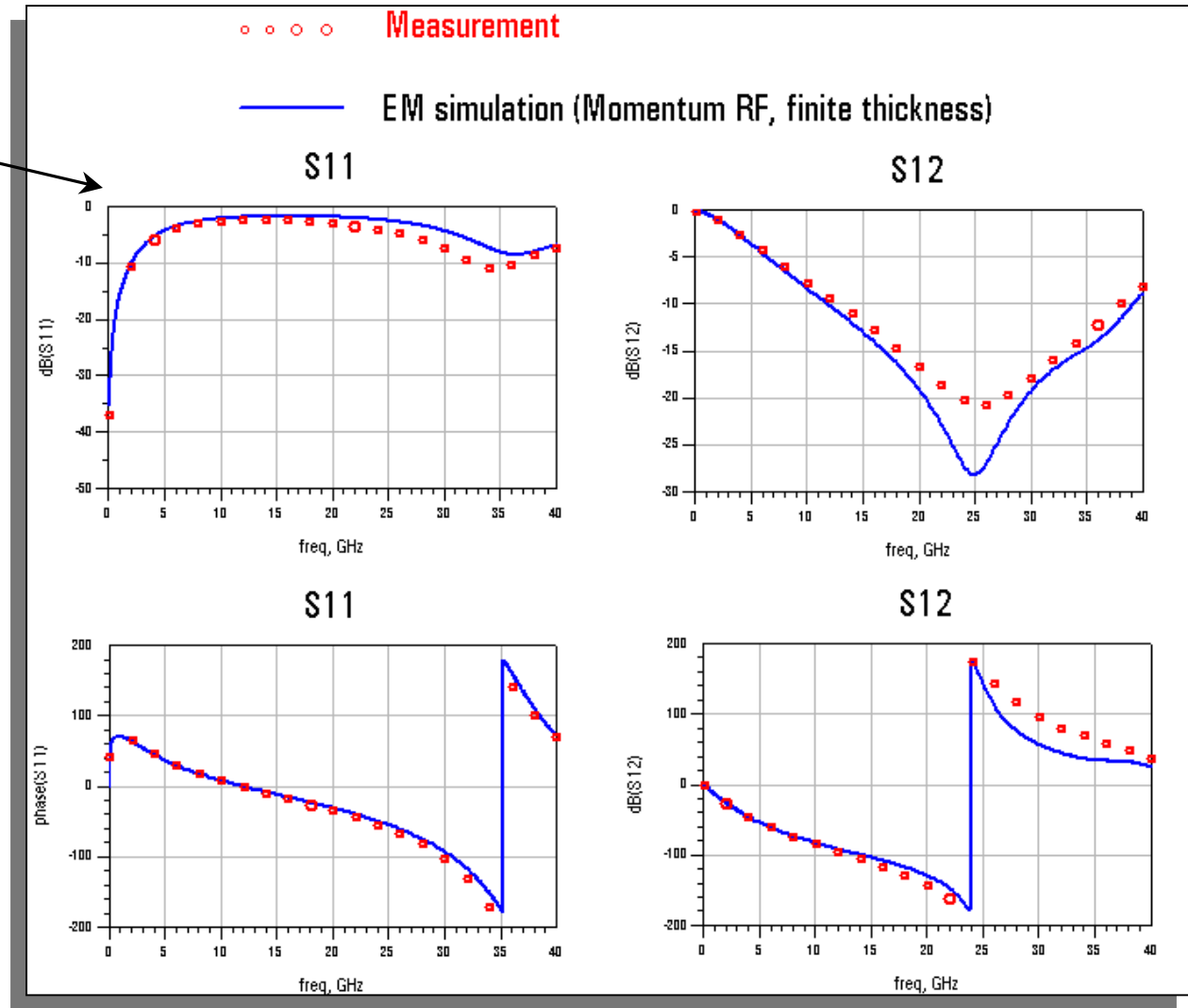
Step 4 – Define Simulation Options

- ✓ Frequency plan
- ✓ Mesh settings
- ✓ Metallization properties...



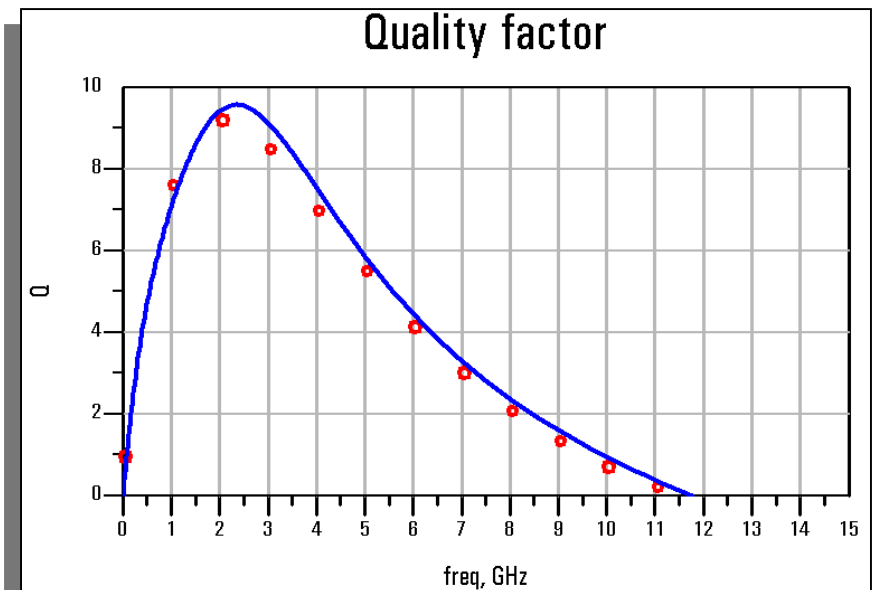
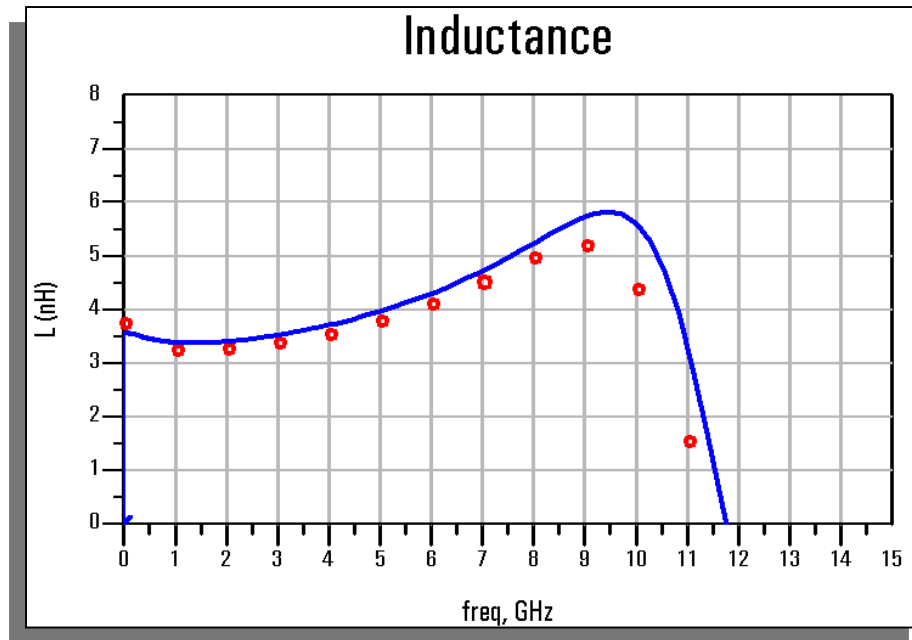
RFDE Momentum Solution Process

Step 5 – Perform EM Simulation & Review Results



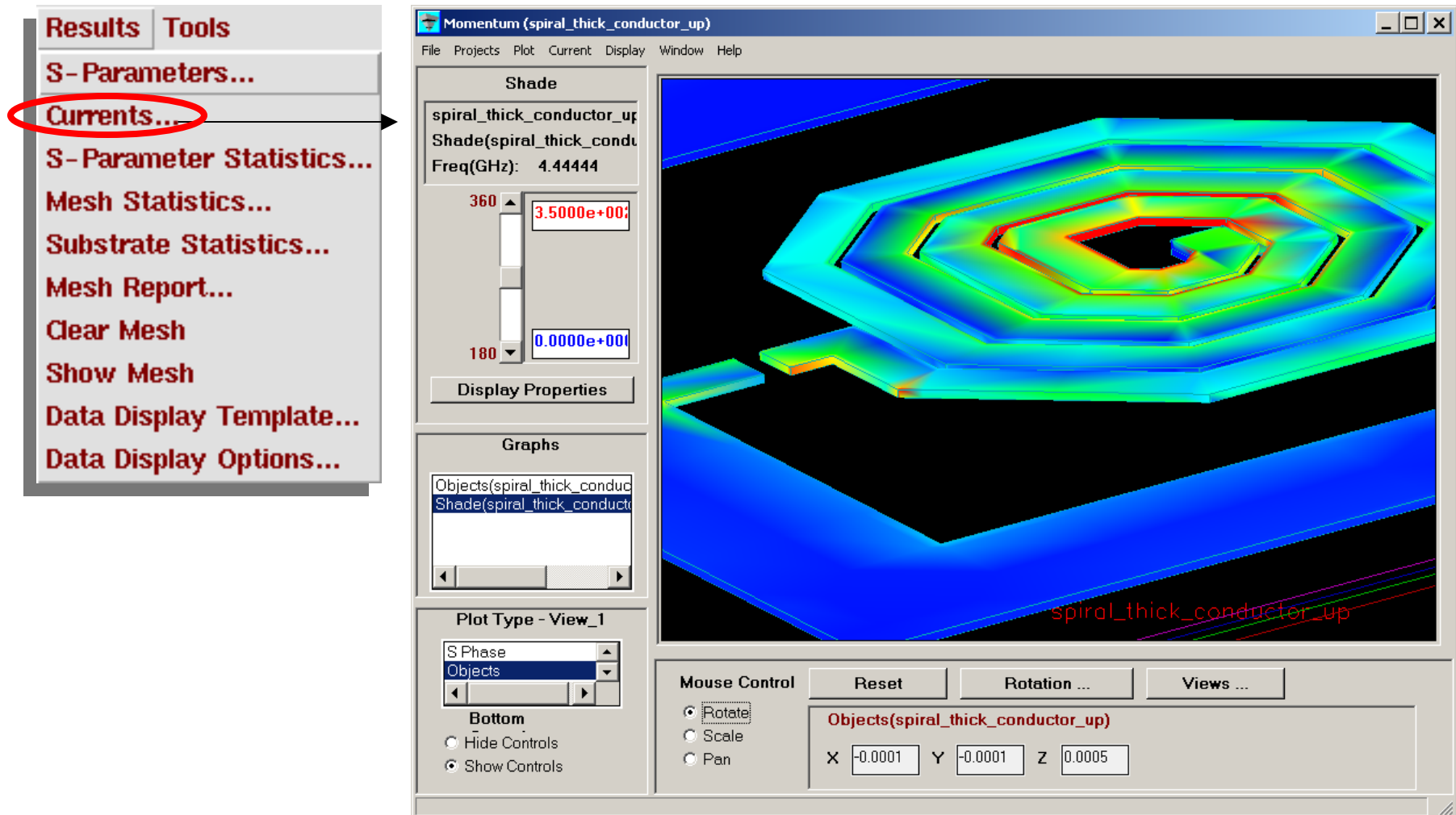
RFDE Momentum Solution Process

Step 5 – Perform EM Simulation & Review Results



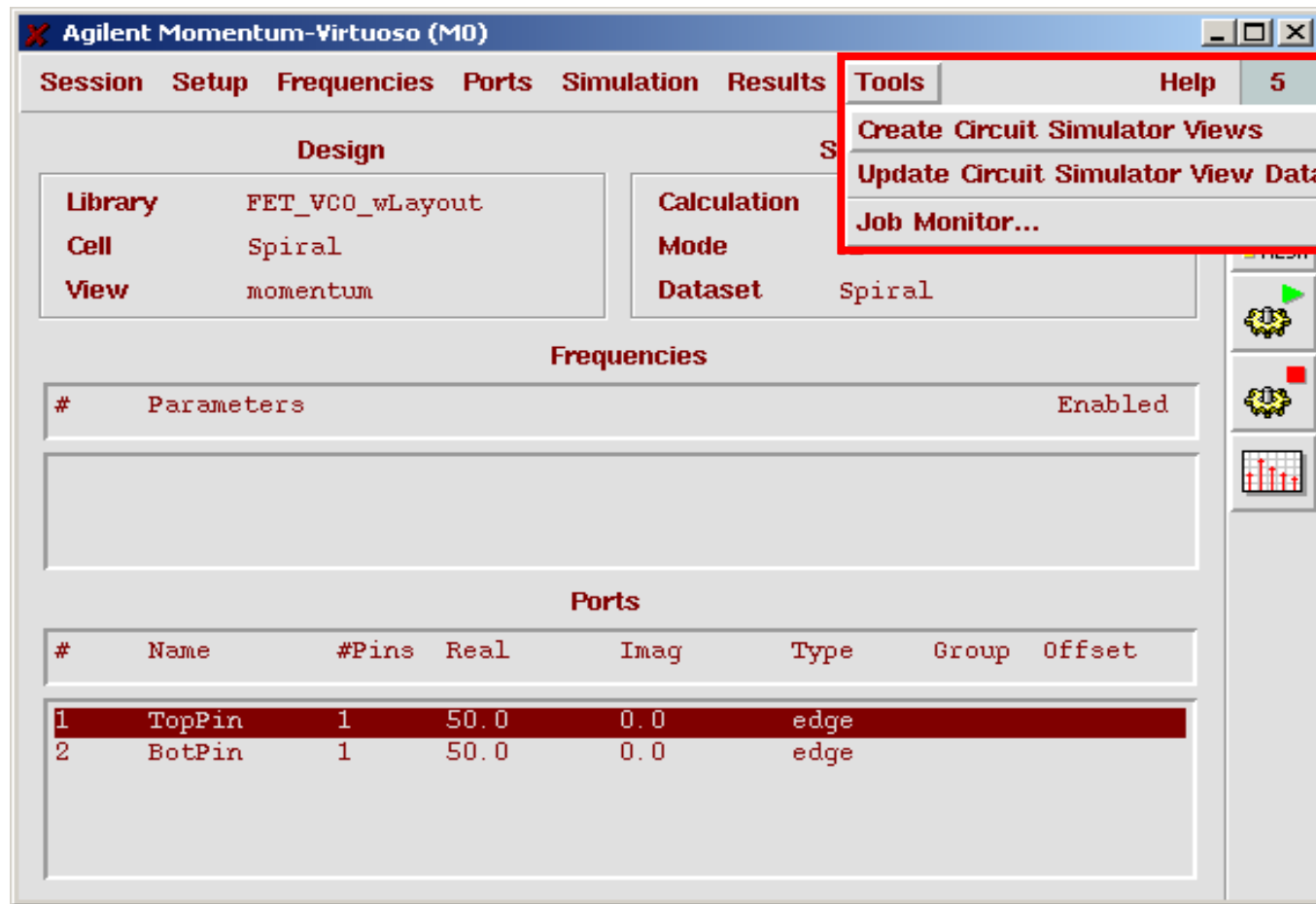
RFDE Momentum Solution Process

Step 5 – Perform EM Simulation & Review Results



RFDE Momentum Solution Process

Step 6 – Make Schematic View for New Model



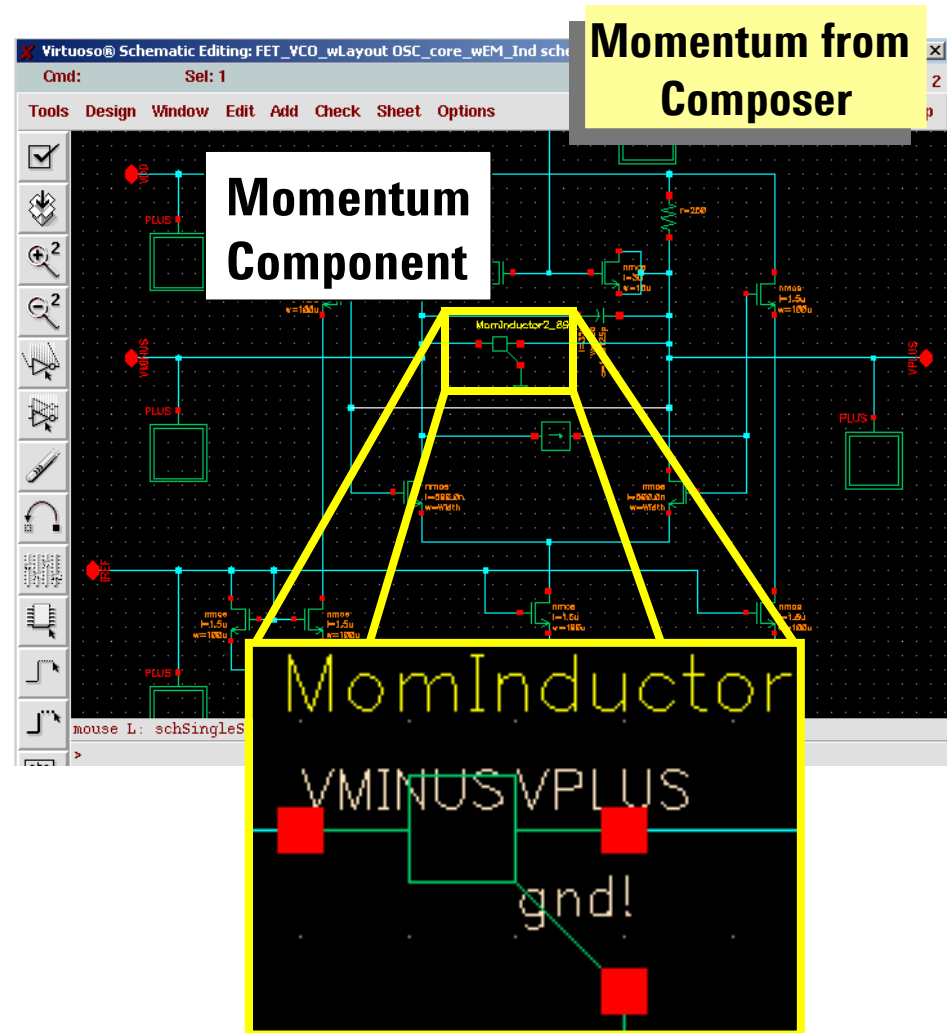
RFDE Momentum Solution Process

Step 6 – Transpose New EM Model to Schematic

Schematic Actions:

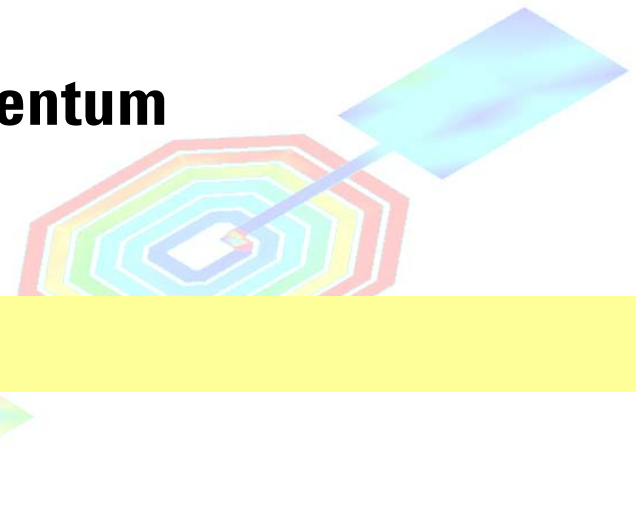
1. Insert Momentum components in Composer
2. Set/Alter simulation control: identical to case where no Momentum components are used
3. Run circuit simulation
4. View/Analyze results

Can be used with Agilent circuit simulator and Cadence Spectre (v5.0.33)



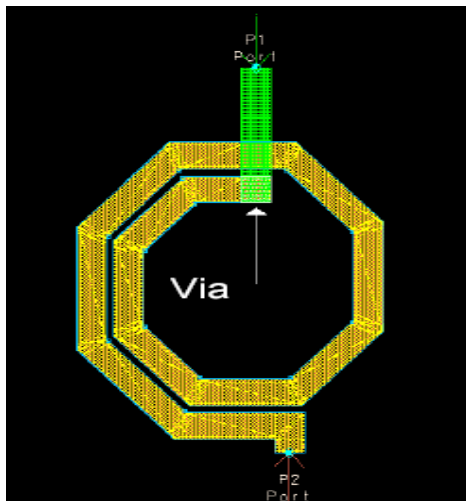
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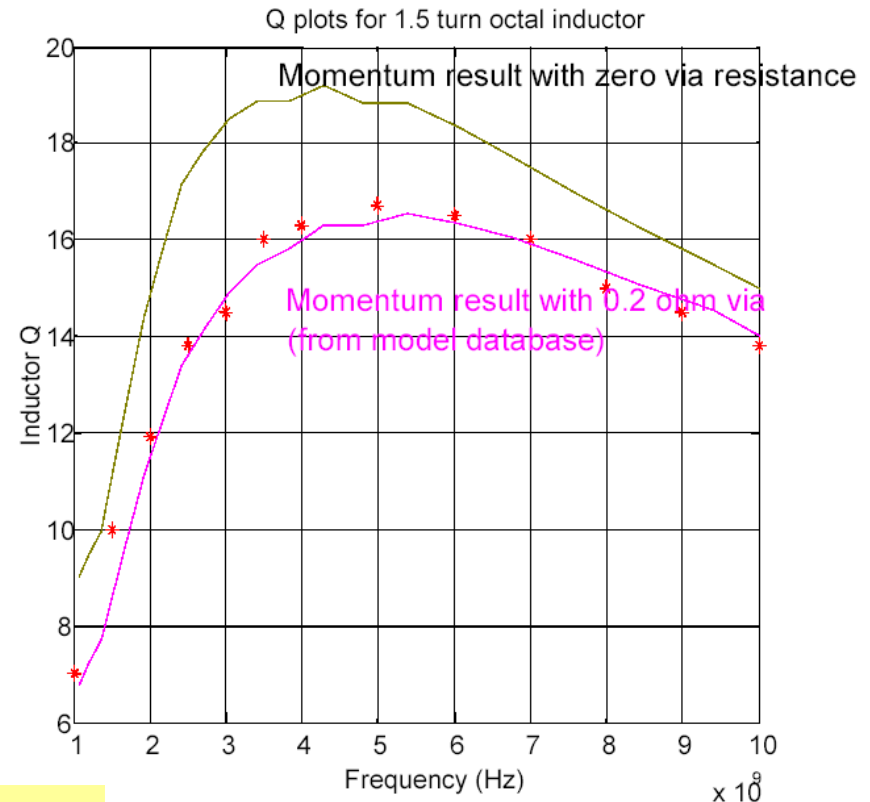


Momentum – Application Benchmark

“Momentum validated against results for standard parts... This leads to a substantial improvement in phase noise of a 4GHz test oscillator” Dr. M P Wilson



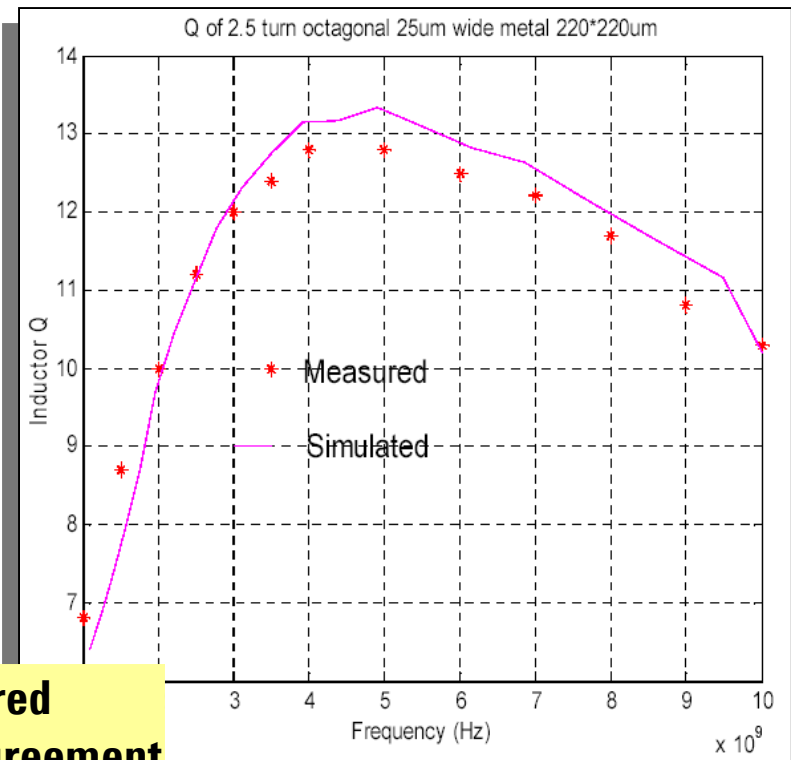
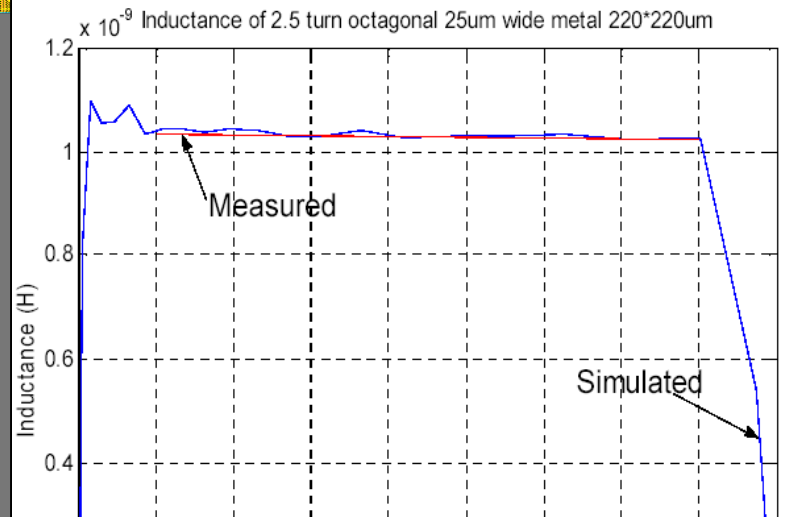
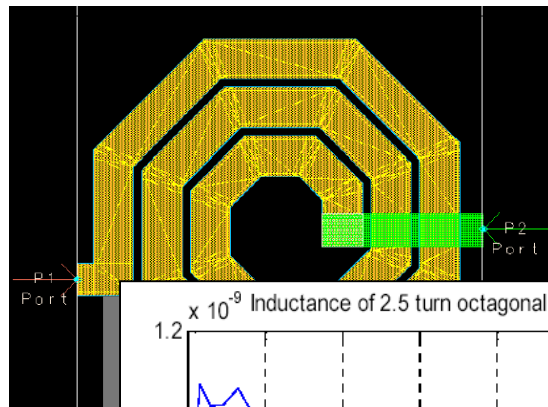
Source: “Modeling of integrated VCO resonators using Momentum” Dr. M P Wilson, Tality UK.



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Momentum – Application Benchmark

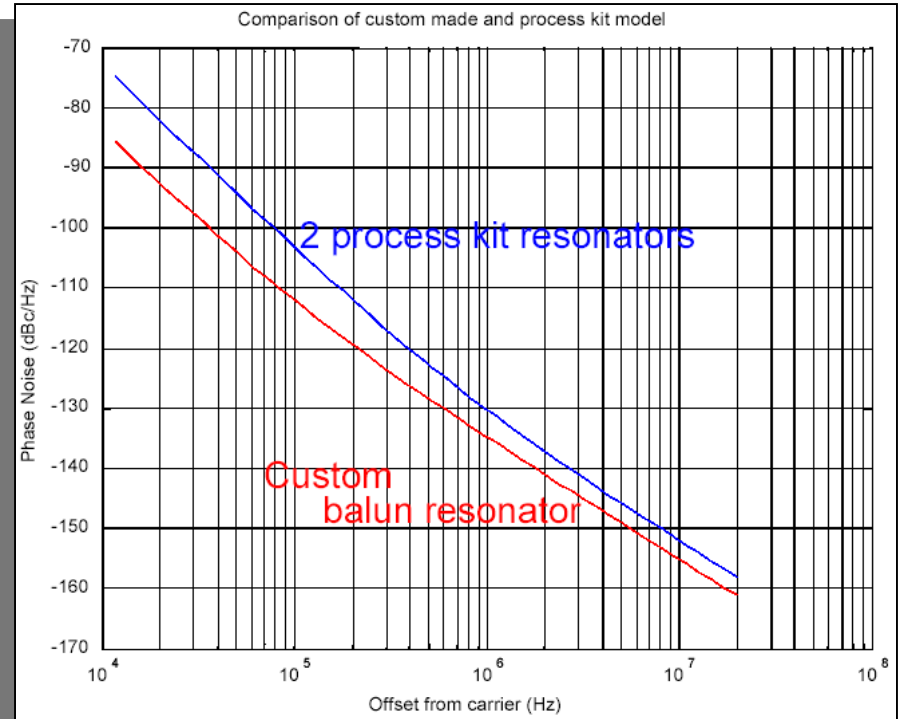
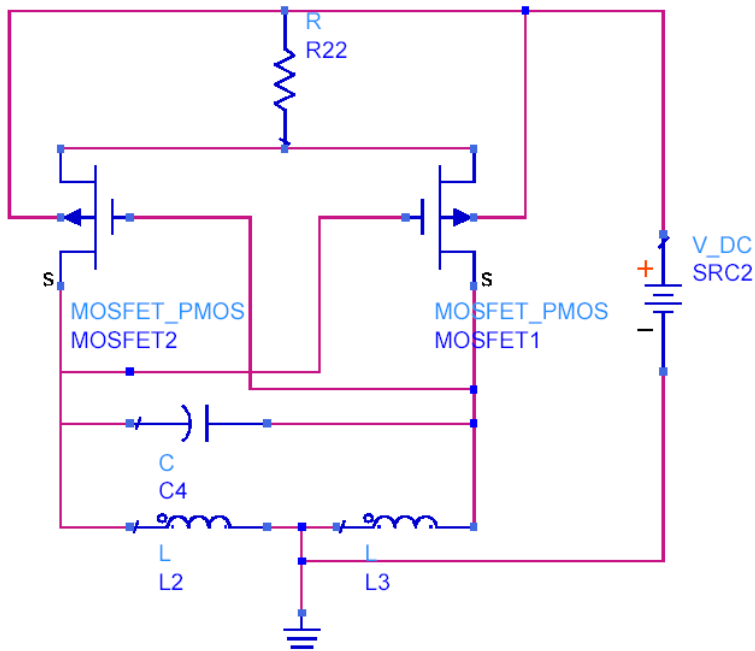
2.5 Turn Octagonal Spiral Inductor



“Three test inductors were compared with measured results. All three inductor examples show good agreement with measured results, showing that we could produce reliable Momentum models for the particular process used.” Dr. M P Wilson.



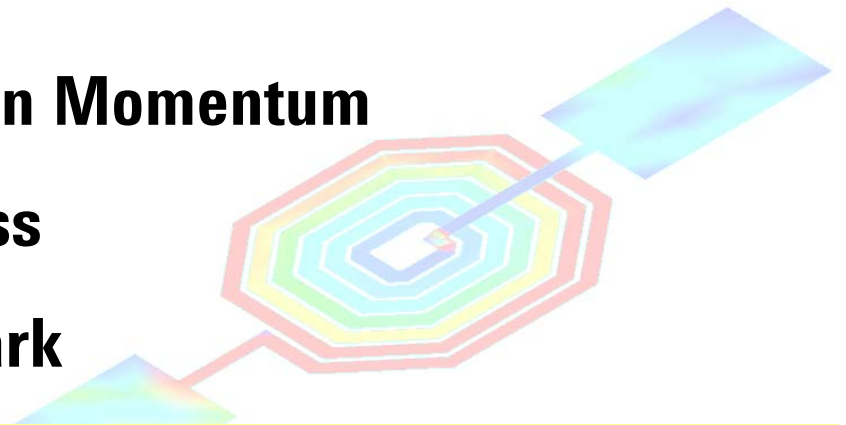
Momentum – Application Benchmark



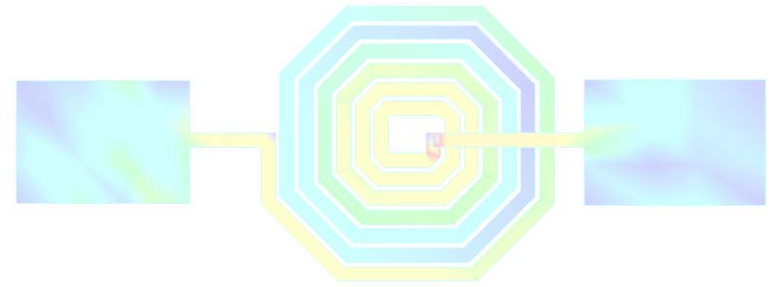
"It can be seen that the Balun circuit improves phase noise by means of an increase in the effective Q of the resonator. This clearly shows the benefit of the custom modeling facility given by Momentum." Dr. M P Wilson

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- **Spiral inductors are critical RFIC design components**
- **Foundries' supplied models may limit design options**
- **Electromagnetic-based modeling approach has many advantages**
- **Method-of-Moments EM technique is best suited for modeling spirals**
- **Momentum industry track record**
- **RFDE Momentum solution process is a simple few steps process**
- **RFDE Momentum is fully integrated with Cadence Virtuoso**
- **RFDE Momentum added accuracy greatly enhances first-pass design success**



Appendix

- **Momentum Substrate Stack Definition**
- **Modeling Metal Losses**



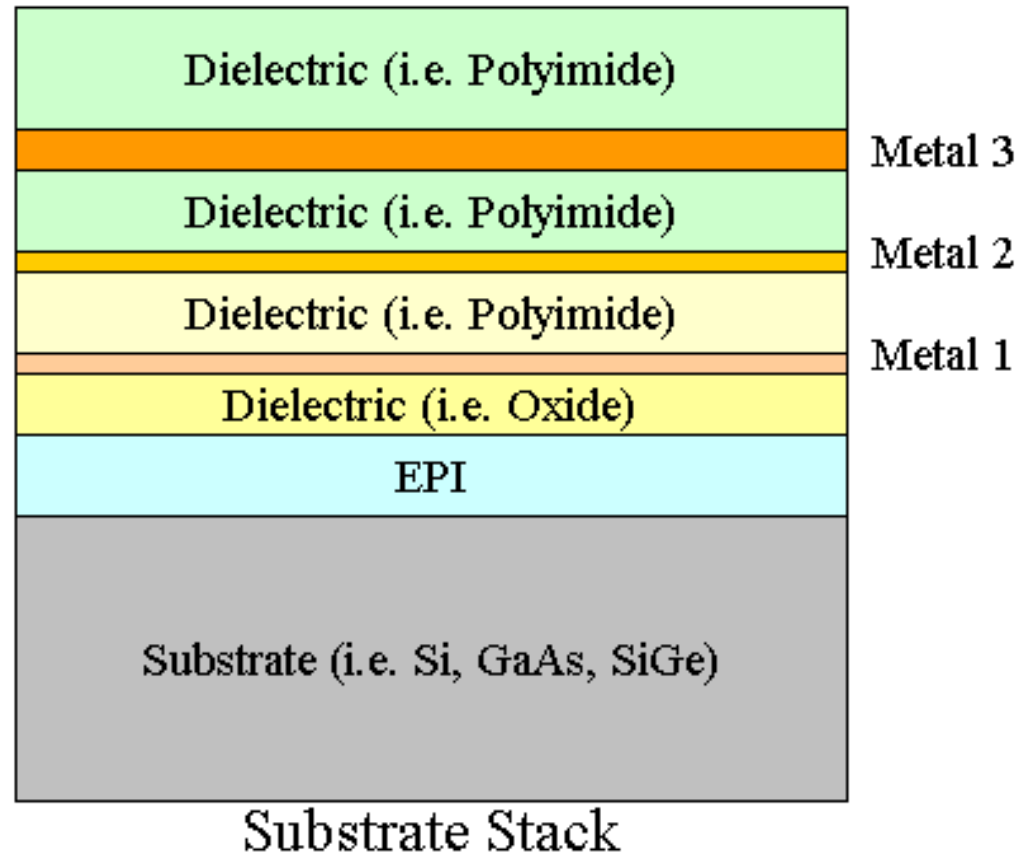
Momentum Substrate Definition

Momentum uses a text file for the description of the substrate definition, which includes:

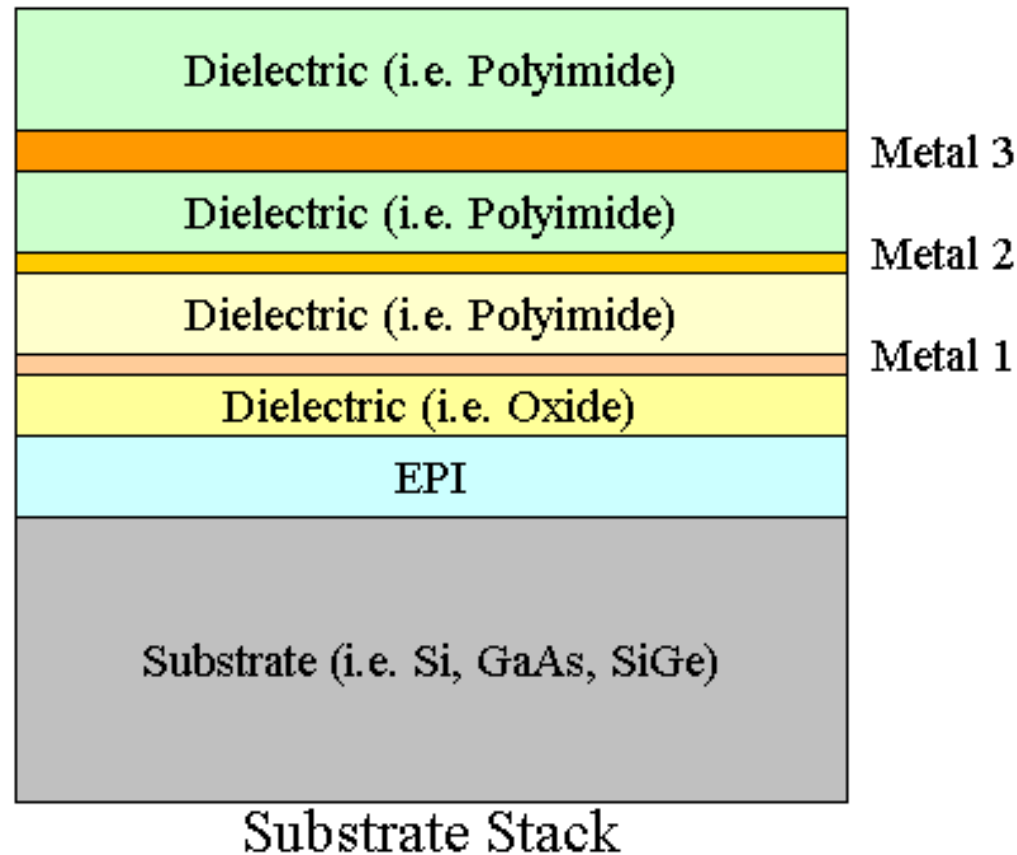
- **Layers stack definition**
- **Dielectric layers physical properties**
- **Metal layers physical properties**



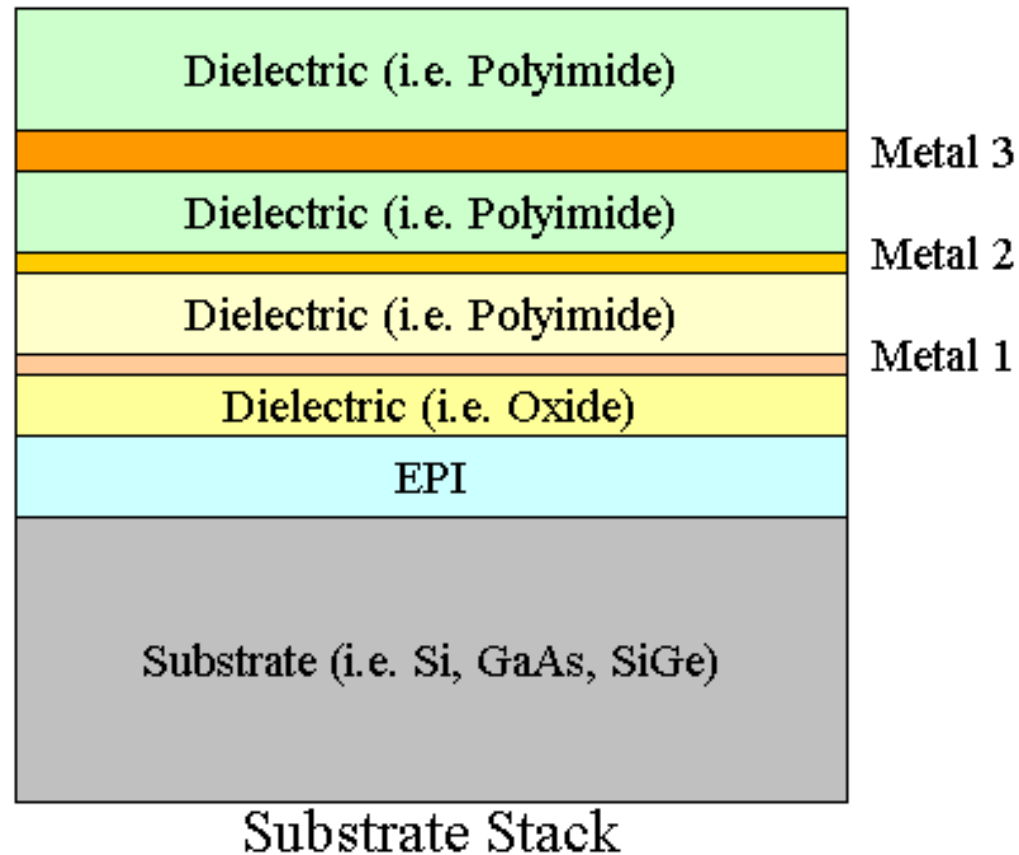
1. Layers Stack Definition



2. Dielectric Layers' Physical Properties



3. Metal Layers Physical Properties



Momentum Substrate Definition – Example

UNITS

DISTANCE = M
 RESISTANCE = OHM/SQ
 CONDUCTIVITY = SIEMENS/M
 RESISTIVITY = OHM.CM
 PERMEABILITY = RELATIVETOVACUUM
 PERMITTIVITY = RELATIVETOVACUUM

END_UNITS

BEGIN_STACK

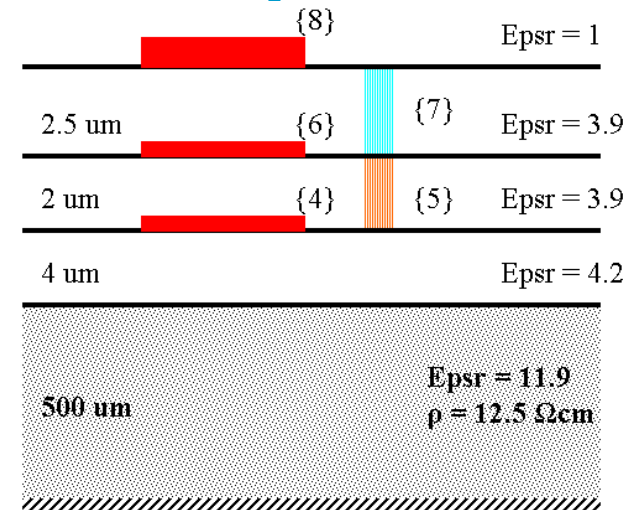
TOP OPEN PERMITTIVITY = 1 PERMEABILITY = 1
 METAL STRIP_MASK = {8}
 DIELECTRIC VIA_MASK = {7} HEIGHT = 2.5e-06 PERMITTIVITY = 3.9 PERMEABILITY = 1
 METAL STRIP_MASK = {6}
 DIELECTRIC VIA_MASK = {5} HEIGHT = 2.0e-06 PERMITTIVITY = 3.9 PERMEABILITY = 1
 METAL STRIP_MASK = {4}
 DIELECTRIC HEIGHT = 4.0e-06 PERMITTIVITY = 4.2 PERMEABILITY = 1
 METAL ABSENT
 DIELECTRIC HEIGHT = 0.0005 PERMITTIVITY = 11.9 PERMEABILITY = 1 RESISTIVITY = 12.5
 BOTTOM COVERED RESISTANCE = 0

END_STACK

BEGIN_MASK

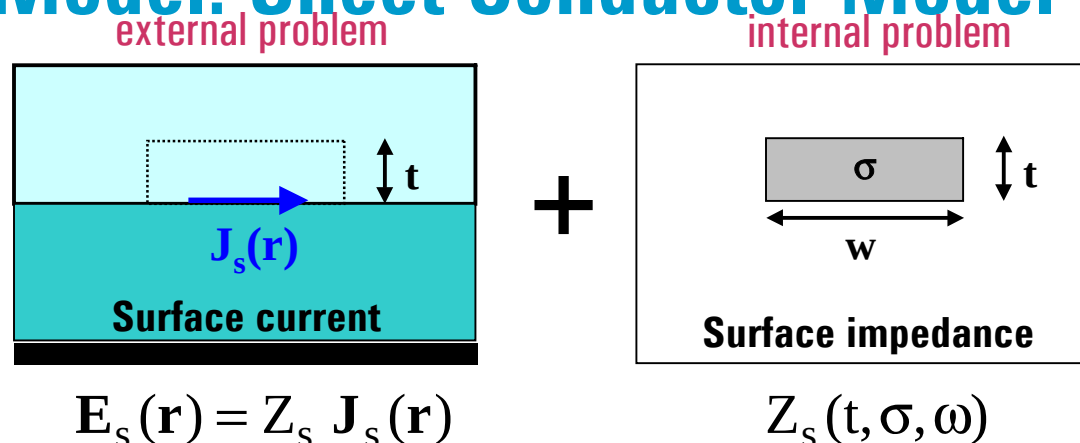
MASK 4 PRECEDENCE = 1 CONDUCTIVITY = 2.7e+07 THICKNESS = 2e-06
 MASK 6 PRECEDENCE = 0 CONDUCTIVITY = 2.7e+07 THICKNESS = 2e-06
 MASK 8 PRECEDENCE = 0 CONDUCTIVITY = 2.7e+07 THICKNESS = 4e-06
 MASK 5 PRECEDENCE = 0 RESISTANCE = 0
 MASK 7 PRECEDENCE = 0 RESISTANCE = 0

END_MASK



Modeling Metal Losses

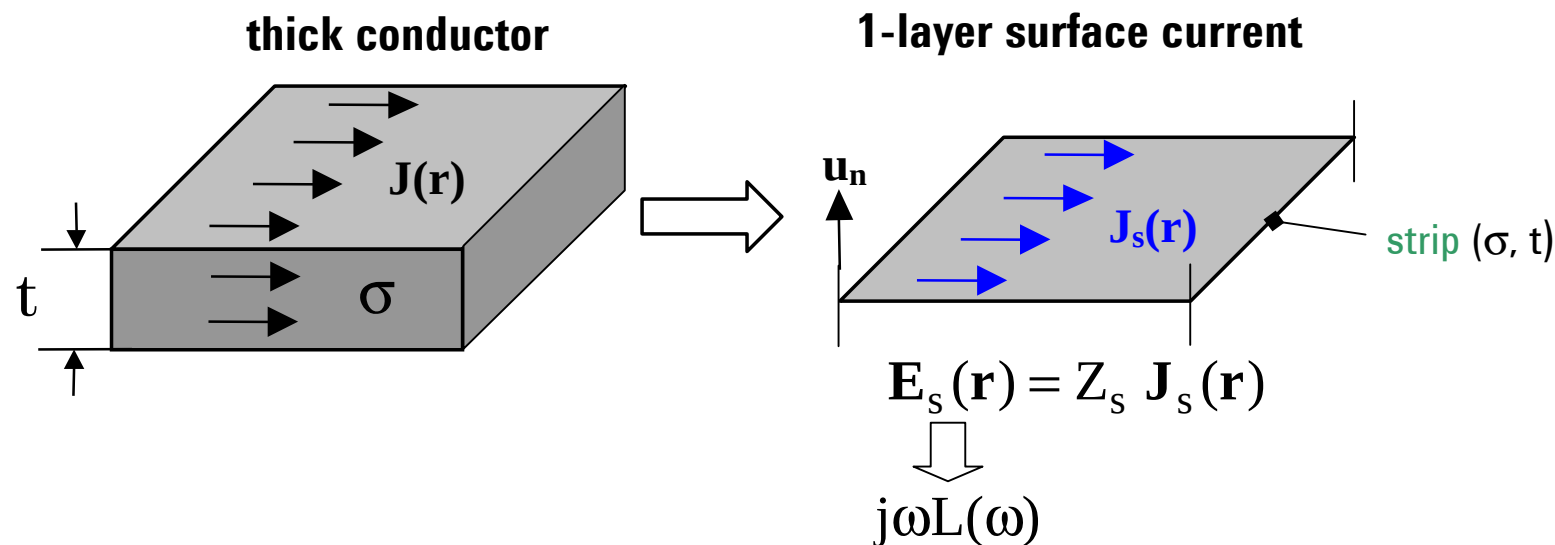
Default Model: Sheet Conductor Model



- Surface impedance based on **1-dimensional field approximation (only variation in z-direction)**, valid for good conductors ($\sigma > \omega\epsilon$) with a high width/thickness ratio (typically $w/t > 5$)
- Models a uniform current distribution over entire cross section at low frequencies, yielding the **correct DC resistance** value
- Models a concentrated current distribution over skin depth δ_s at high frequencies, **over estimates HF resistance** (depending on closeness of return current in ground plane)
- Models the external inductance for zero thickness with higher internal inductance (skin effect), hence **over estimates inductance**

Modeling Metal Losses

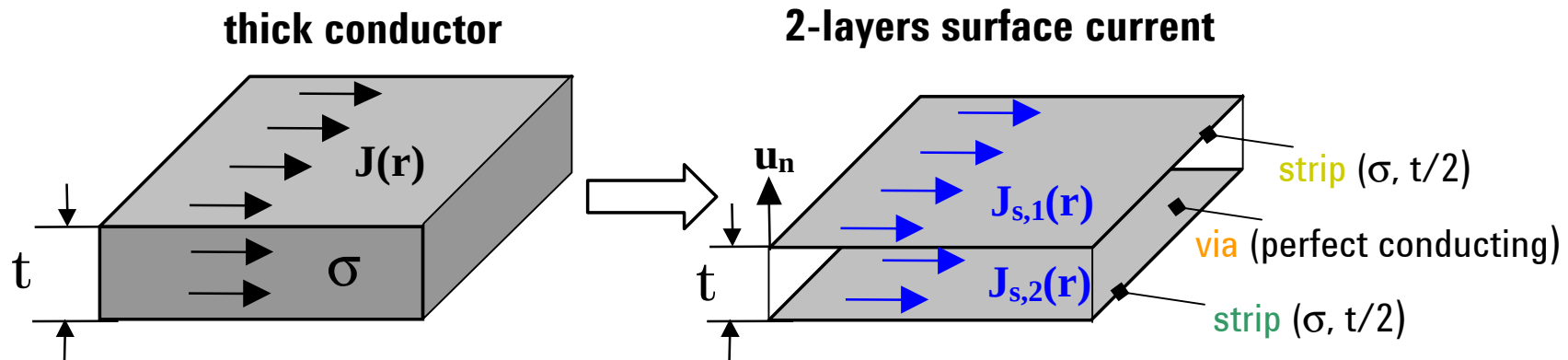
Surface Currents – Sheet Conductors



- Volume of thick conductor is modeled as an infinitely thin conductor layer
- Surface current flows in **one sheet conductor layer**
- External inductance $L(\omega)$ is **independent** of conductor thickness (t)

Modeling Metal Losses

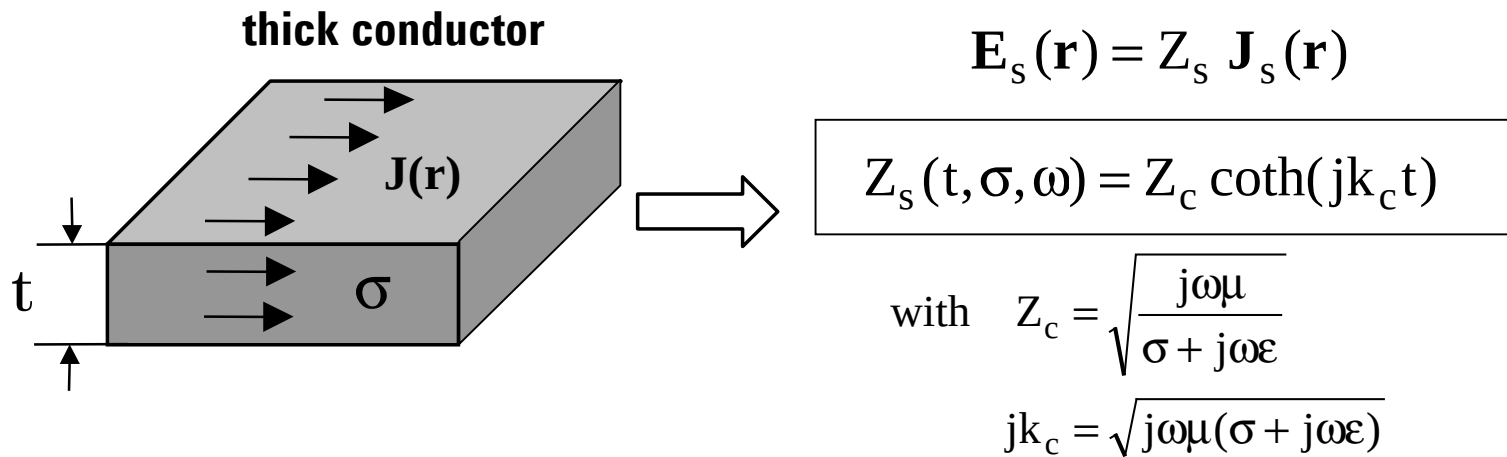
Surface Currents – Thick Conductors



- Volume of thick conductor is divided into two equal conductor layers, **one at the top surface** and **one at the bottom surface**, with same conductivity and half of the conductor thickness
- Additional **via layer** (perfect conducting) included to short out differential mode
- Actual distribution of top and bottom layer currents not enforced, but follows from solution of EM equations
- External inductance $L(\omega)$ is **dependent** on conductor thickness
- As conductor thickness (t) increases: mutual inductance decreases $\rightarrow$ external inductance decreases; also, high frequency resistance decreases

Modeling Metal Losses

1-dimensional surface impedance model



- Distribution of the current inside the conductor is based on **1-dimensional field approximation (only variation in z-direction)**, valid for good conductors ($\sigma > \omega\epsilon$) with a high width/thickness ratio (typically $w/t > 5$)
- Yields analytic model for the surface impedance $Z_s(t, \sigma, \omega)$
- Resistance **$R(\omega)$** and interior inductance **$\Delta L(\omega)$** are **dependent** on: conductor thickness, conductivity, and frequency (**skin effect**)

Modeling Metal Losses

1-dimensional surface impedance model

$$Z_s(t, \sigma, \omega) = Z_c \coth(jk_c t)$$

with $Z_c = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$

$$jk_c = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$

low frequencies (LF):

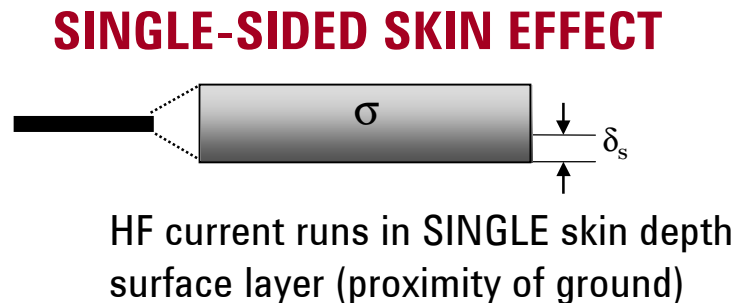
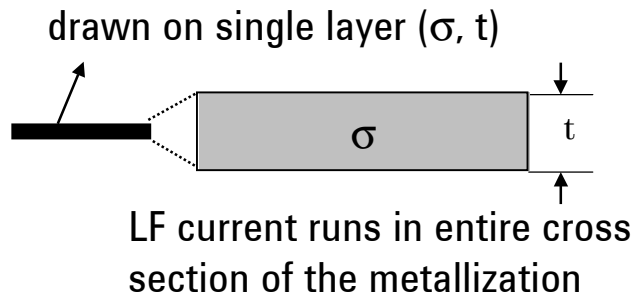
$$Z_s(t, \sigma, \omega) \rightarrow \frac{1}{\sigma t} + j\omega \frac{\mu t}{3}$$

$$R_{DC} + j\omega\Delta L_{DC}$$

high frequencies (HF):

$$Z_s(t, \sigma, \omega) \rightarrow \frac{1}{\sigma\delta_s}(1 + j) \quad \text{skin depth } \delta_s = \sqrt{\frac{2}{\omega\mu\sigma}}$$

$$R_{HF} + j\omega\Delta L_{HF}$$



Typical Application: microstrip circuit

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